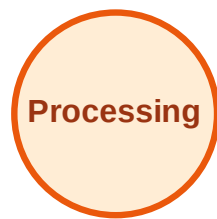


# BFS Traversal

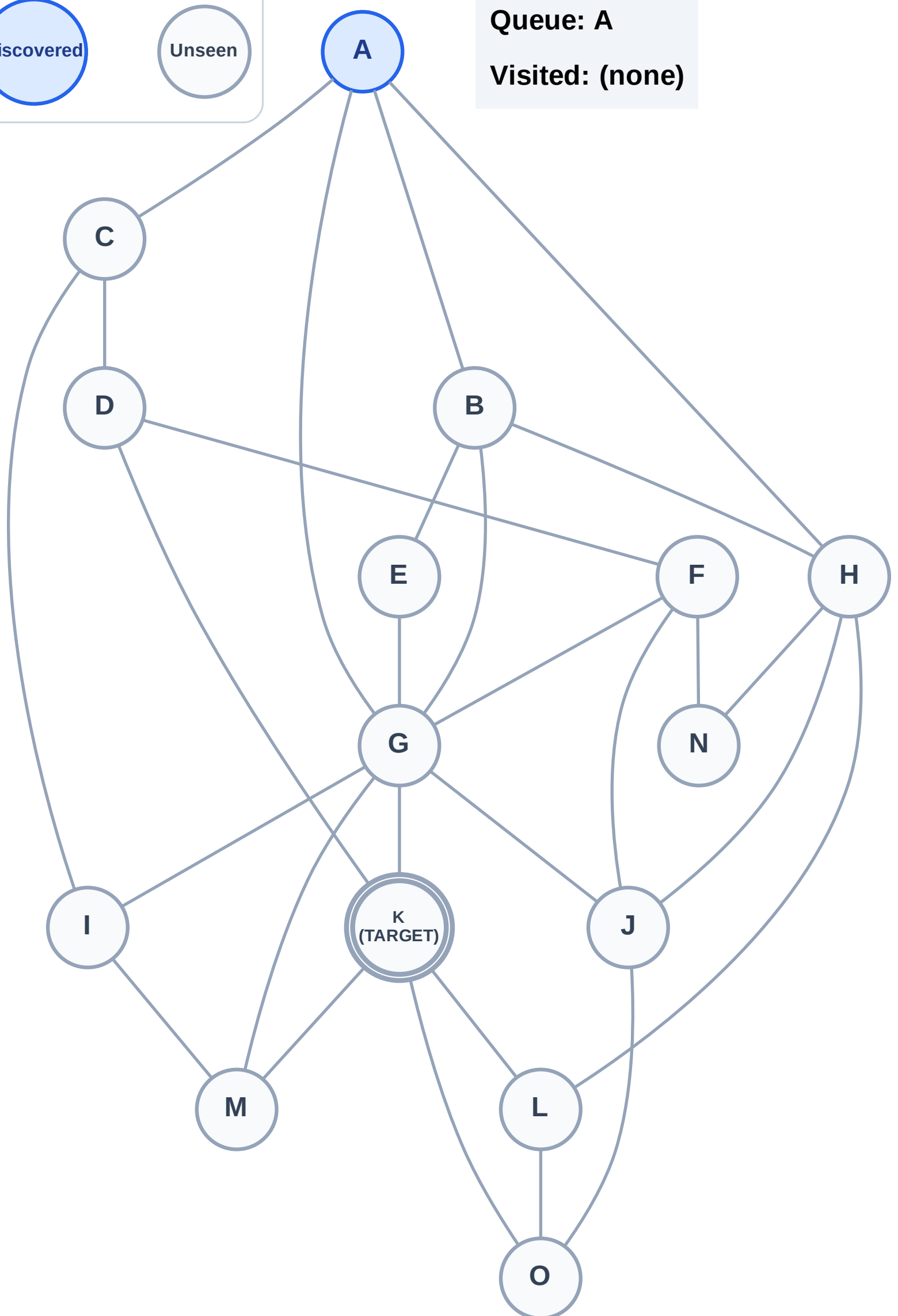
Step 1: Start at A (Seeking target node K)

## Legend



Queue: A

Visited: (none)



# BFS Traversal

Step 2: Process node A

Legend

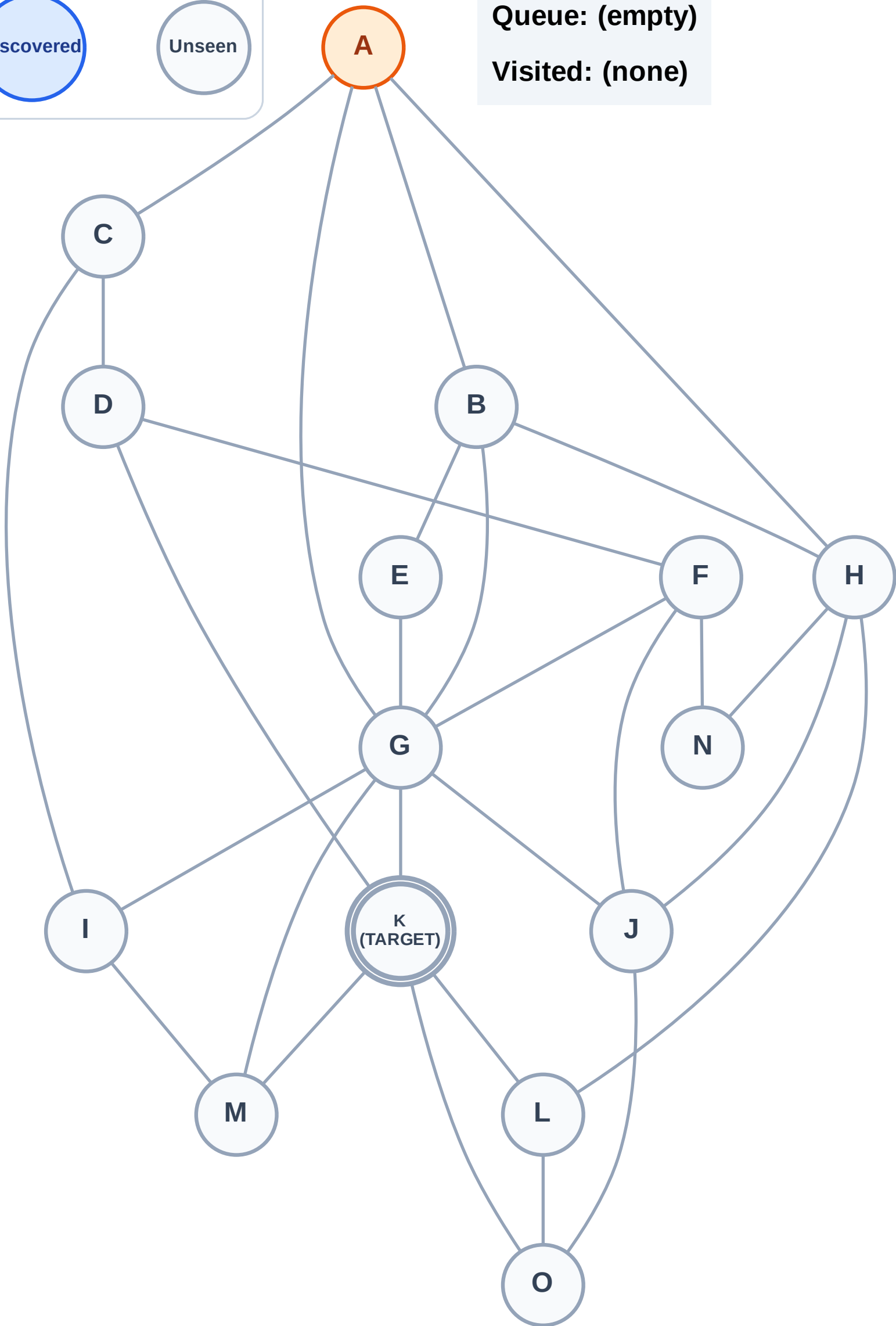
Complete

Processing

Discovered

Unseen

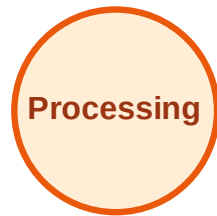
Queue: (empty)  
Visited: (none)



# BFS Traversal

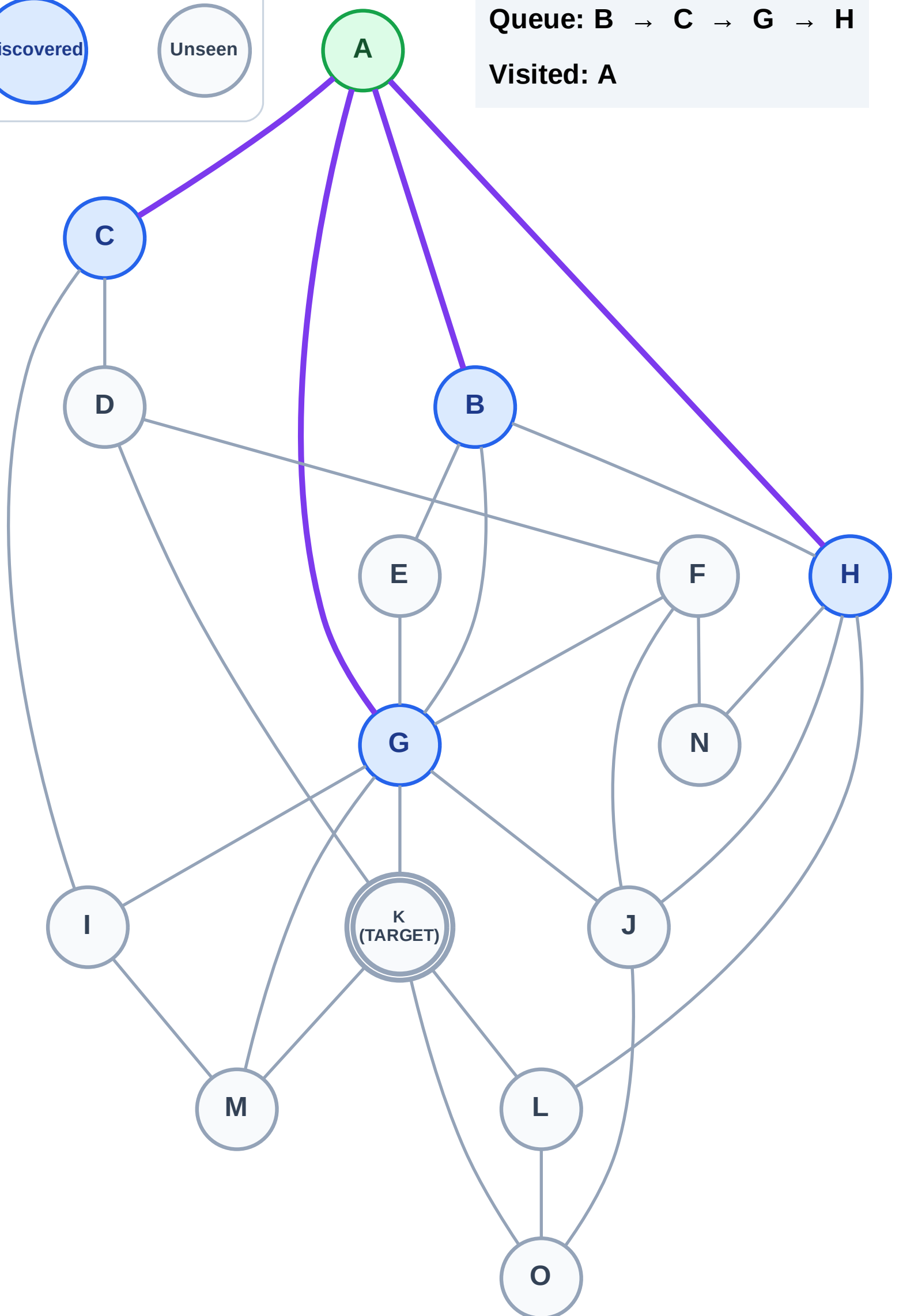
Step 3: Discover B, C, G, H from A

## Legend



Queue: B → C → G → H

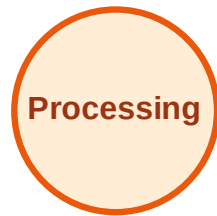
Visited: A



# BFS Traversal

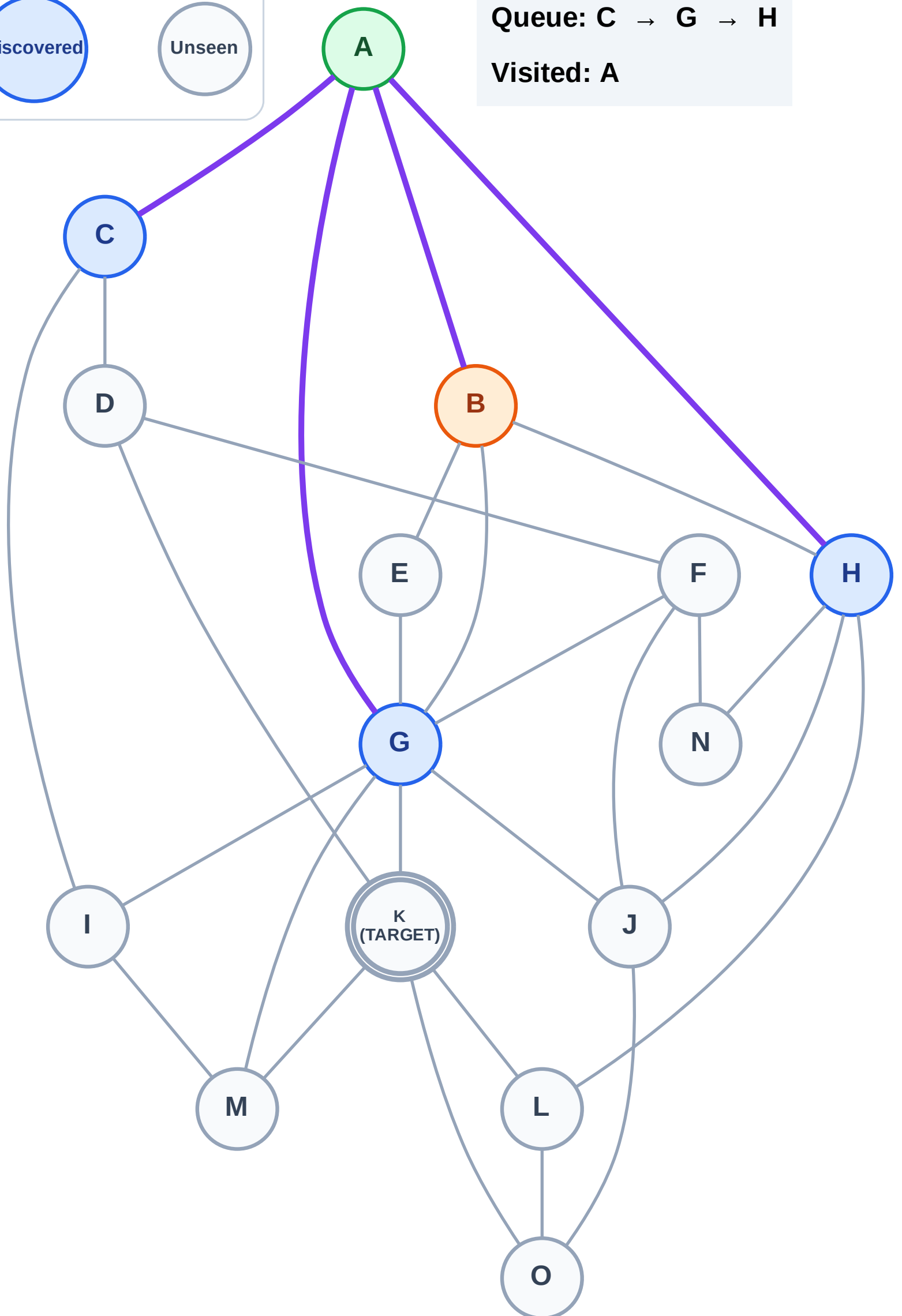
Step 4: Process node B

## Legend



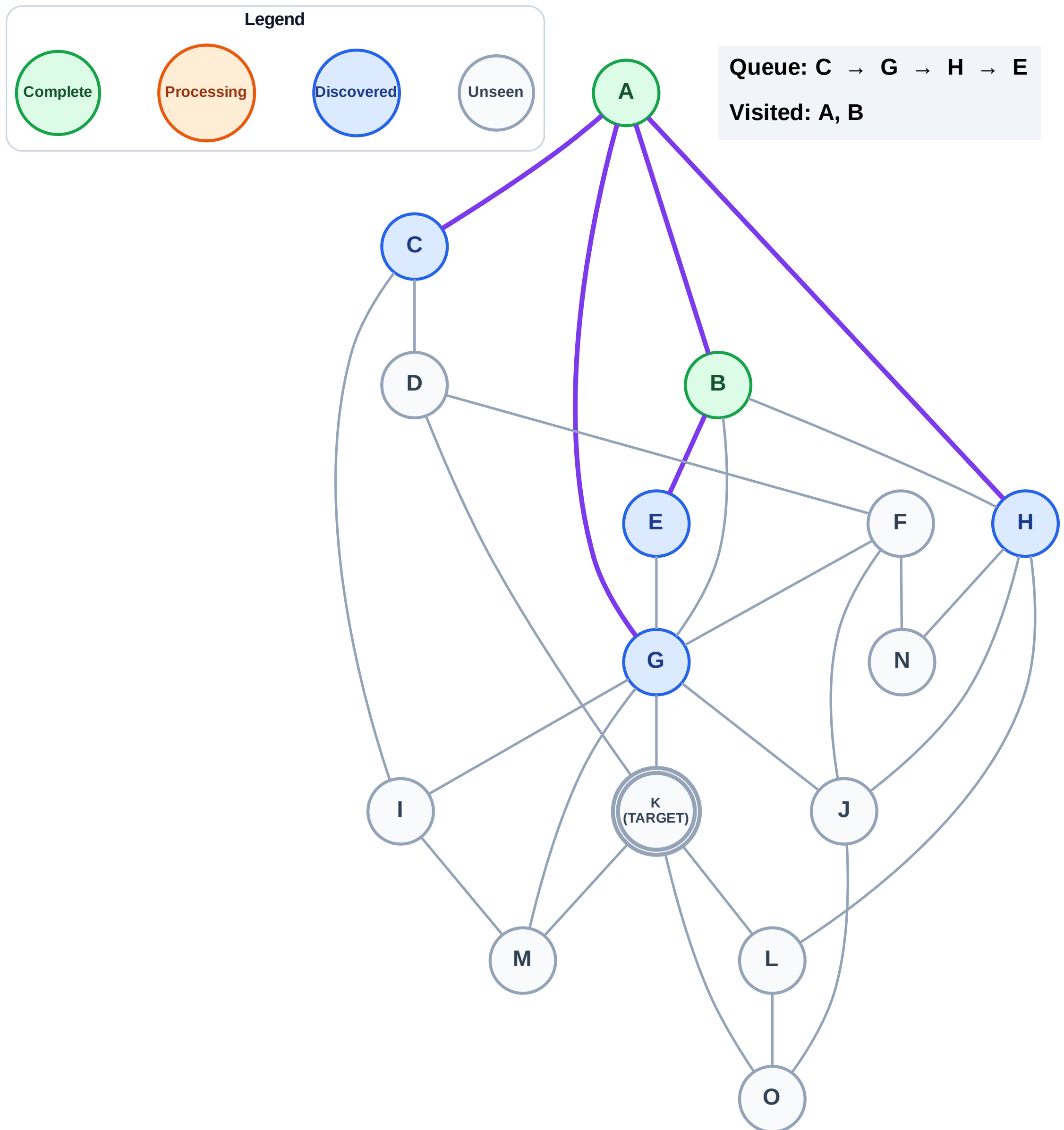
Queue: C → G → H

Visited: A



## BFS Traversal

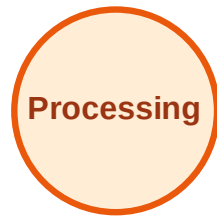
### Step 5: Discover E from B



# BFS Traversal

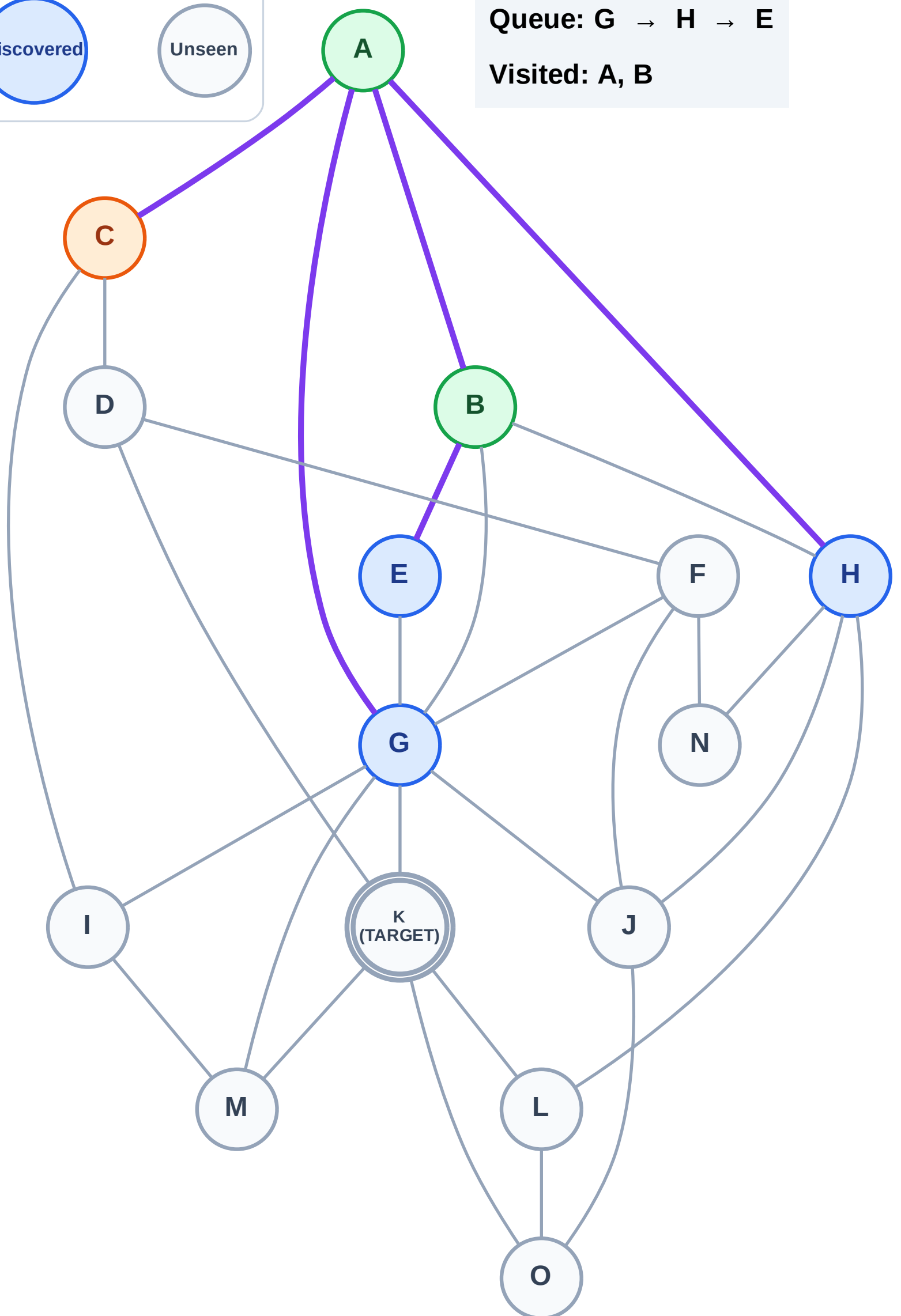
Step 6: Process node C

## Legend



Queue: G → H → E

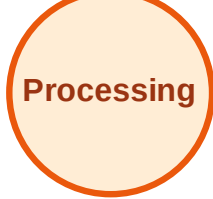
Visited: A, B



# BFS Traversal

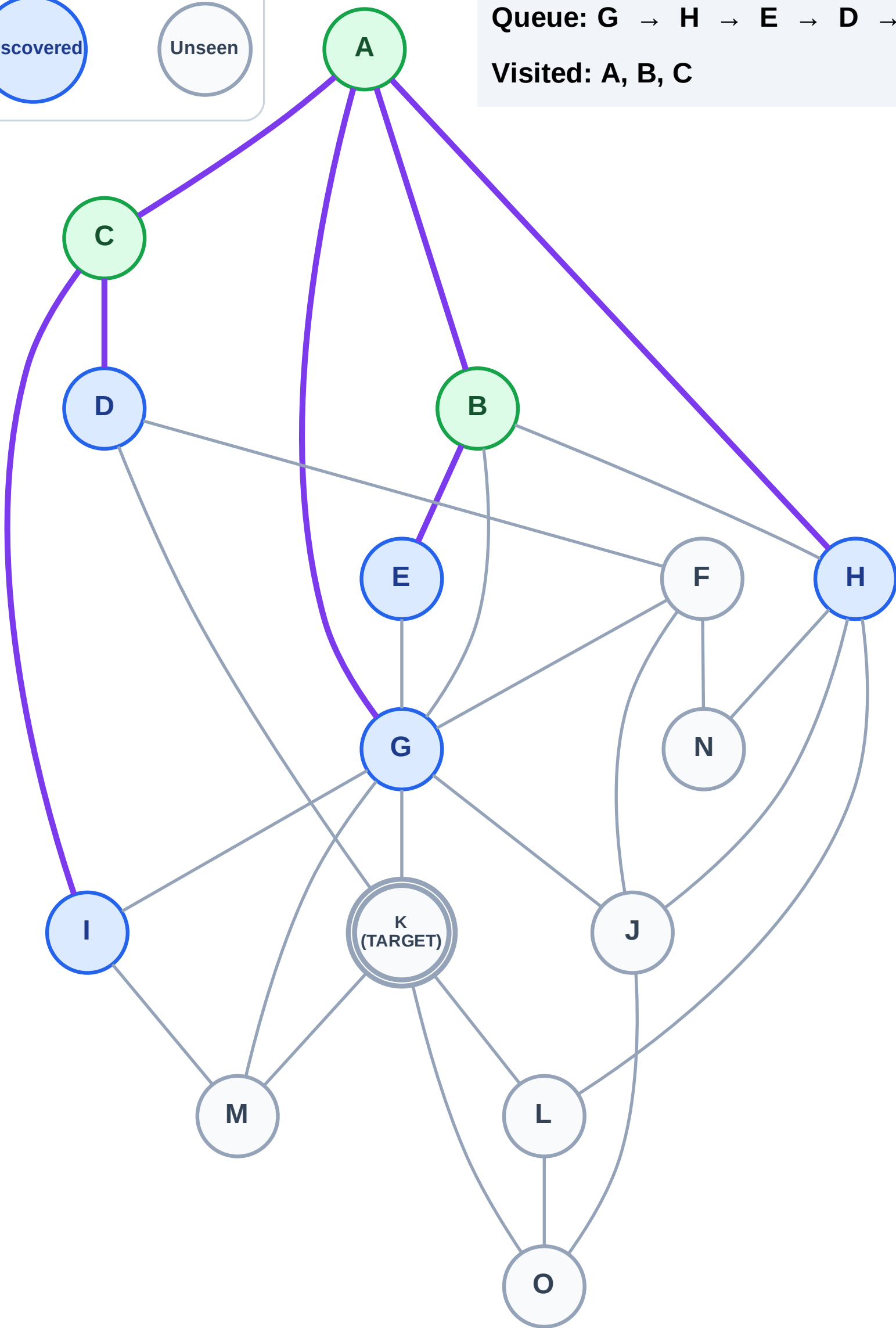
Step 7: Discover D, I from C

## Legend



Queue: G → H → E → D → I

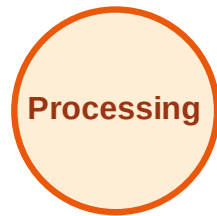
Visited: A, B, C



# BFS Traversal

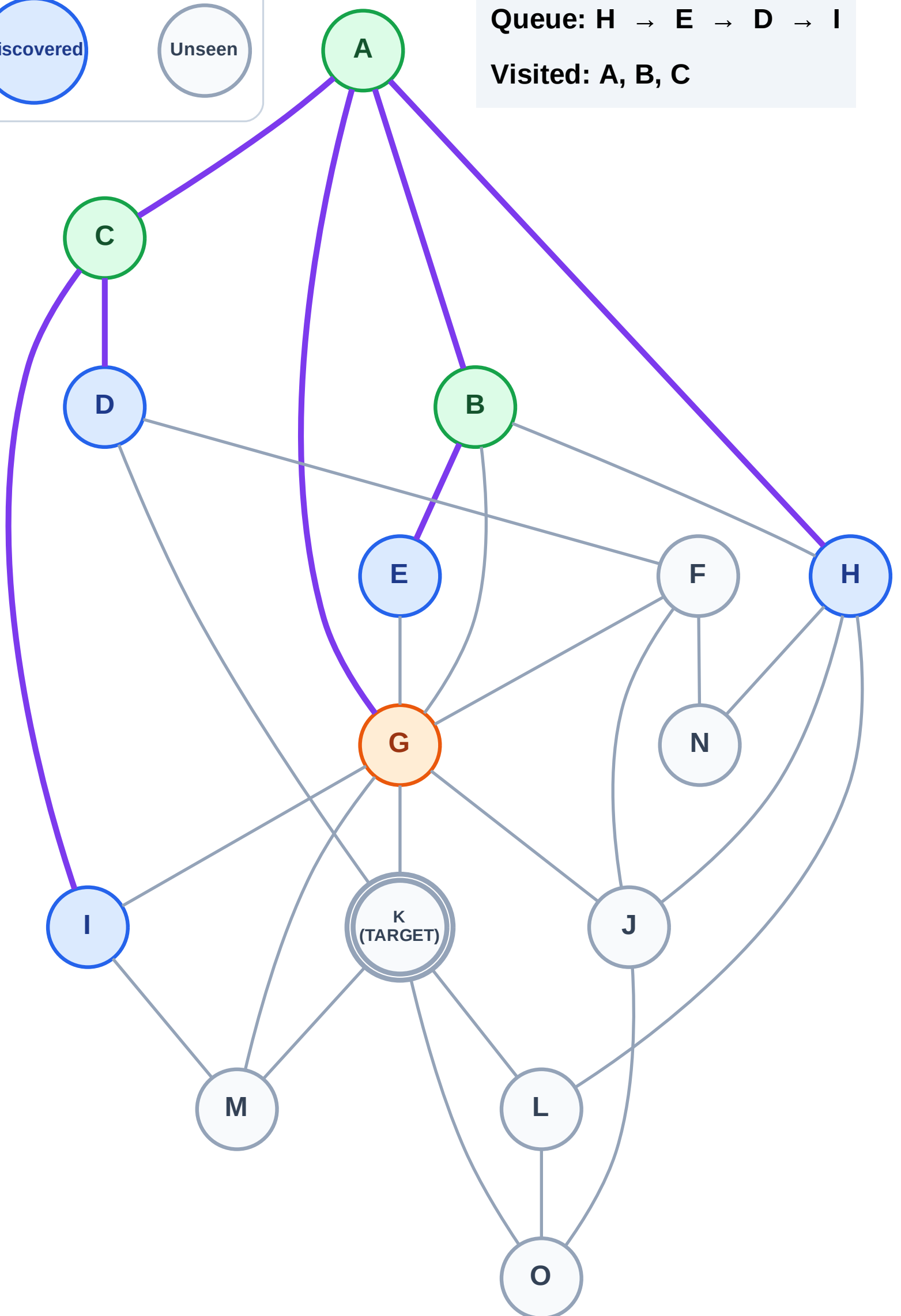
Step 8: Process node G

## Legend



Queue: H → E → D → I

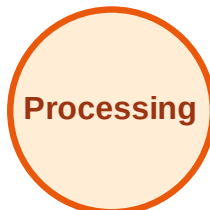
Visited: A, B, C



# BFS Traversal

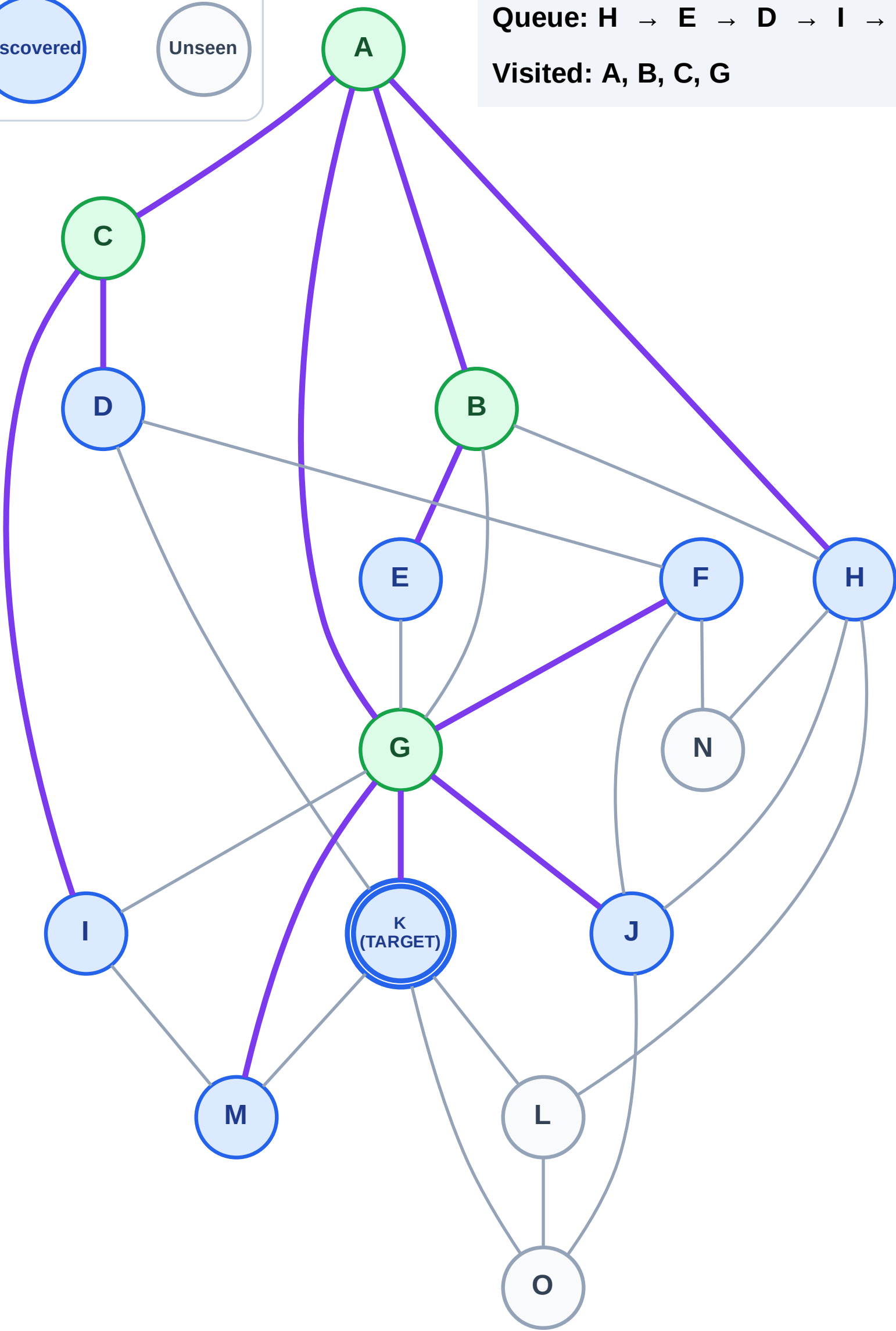
Step 9: Discover F, J, K, M from G

## Legend



Queue: H → E → D → I → F → J → K → M

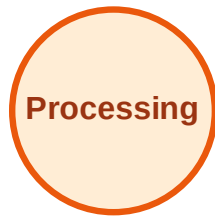
Visited: A, B, C, G



# BFS Traversal

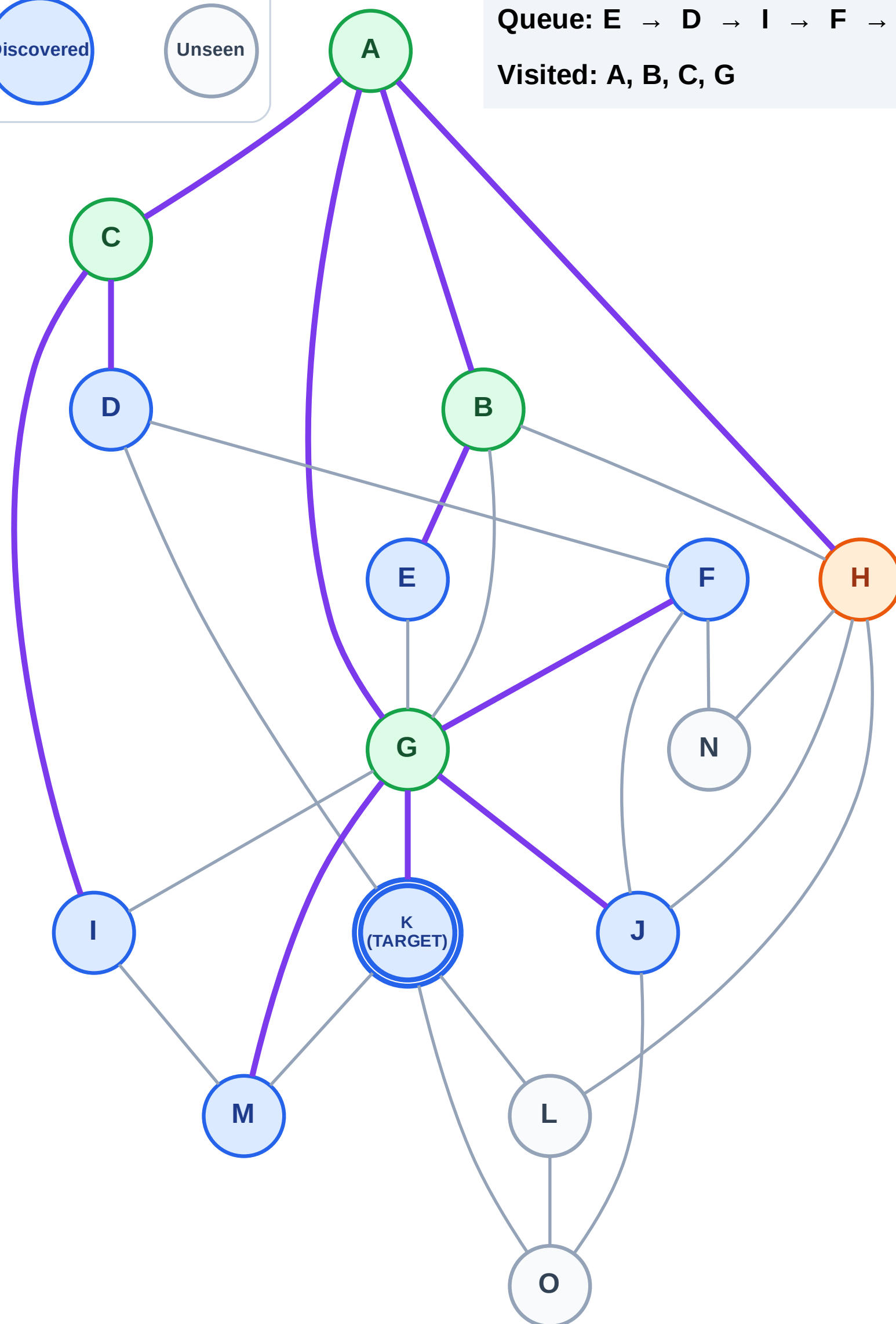
Step 10: Process node H

## Legend



Queue: E → D → I → F → J → K → M

Visited: A, B, C, G



# BFS Traversal

Step 11: Discover L, N from H

## Legend

Complete

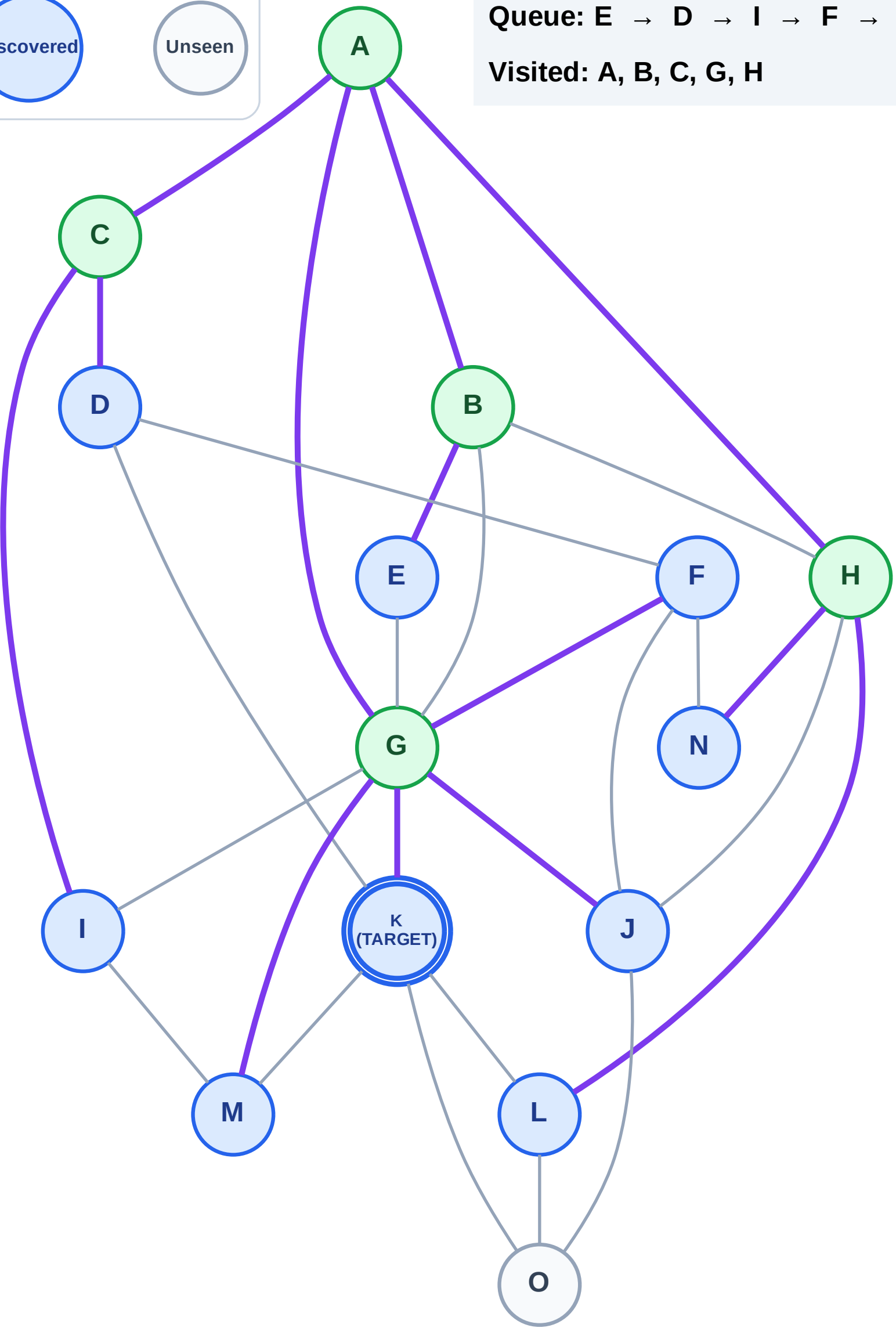
Processing

Discovered

Unseen

Queue: E → D → I → F → J → K → M → L → N

Visited: A, B, C, G, H



# BFS Traversal

Step 12: Process node E

Legend

Complete

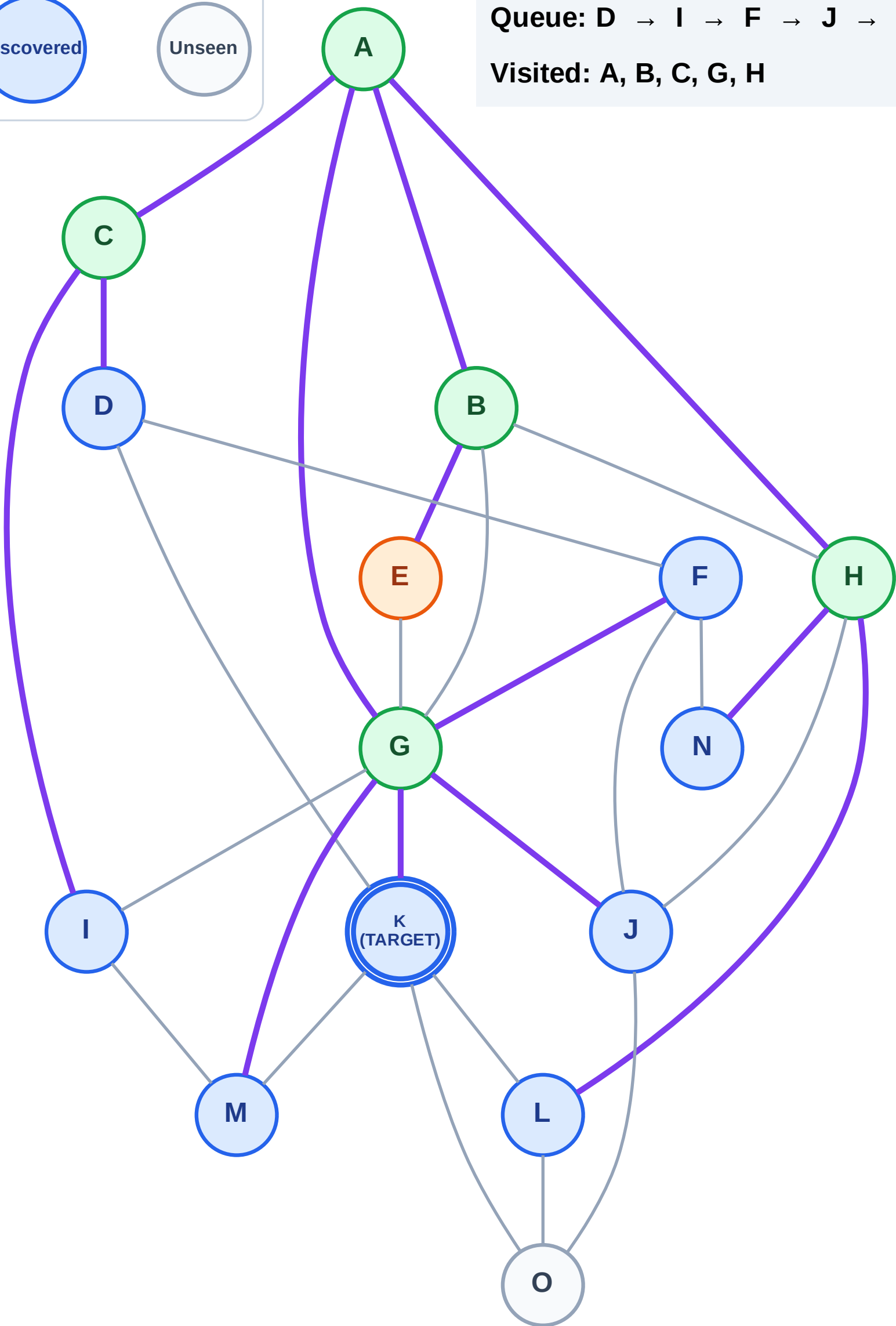
Processing

Discovered

Unseen

Queue: D → I → F → J → K → M → L → N

Visited: A, B, C, G, H



# BFS Traversal

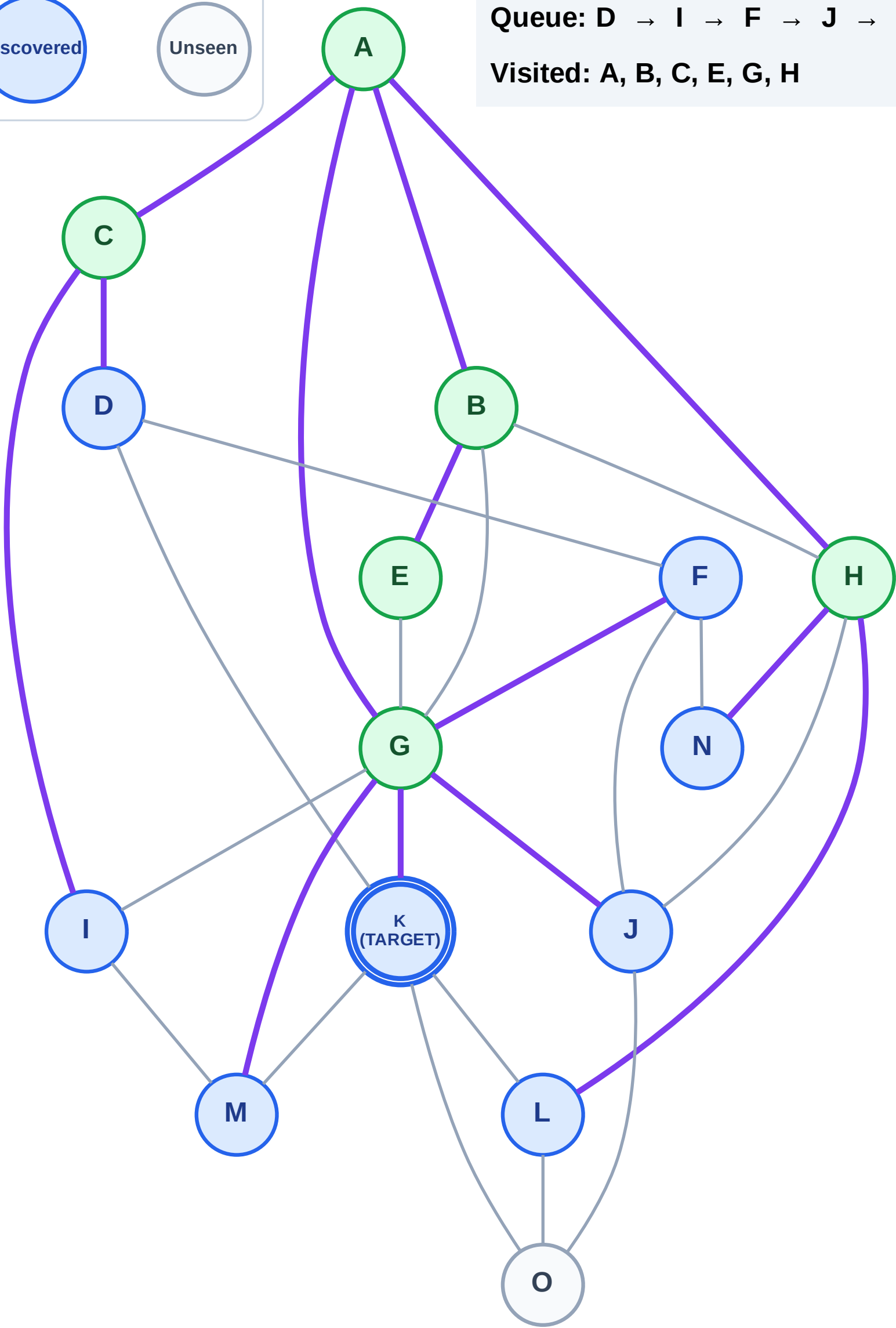
Step 13: No new neighbors from E

## Legend



Queue: D → I → F → J → K → M → L → N

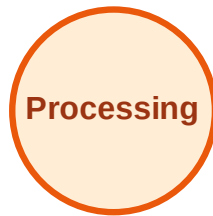
Visited: A, B, C, E, G, H



# BFS Traversal

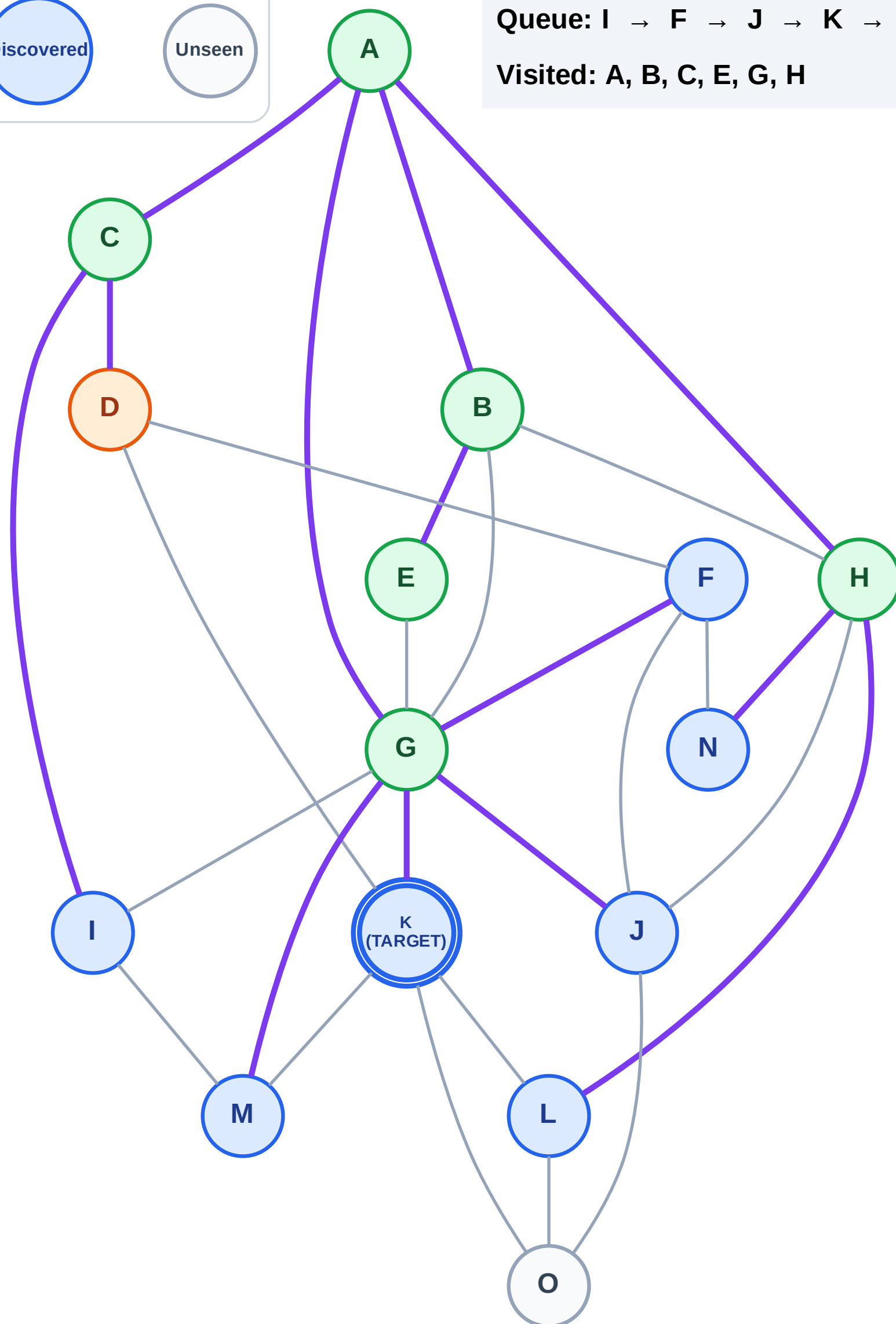
Step 14: Process node D

## Legend



Queue: I → F → J → K → M → L → N

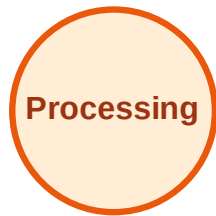
Visited: A, B, C, E, G, H



# BFS Traversal

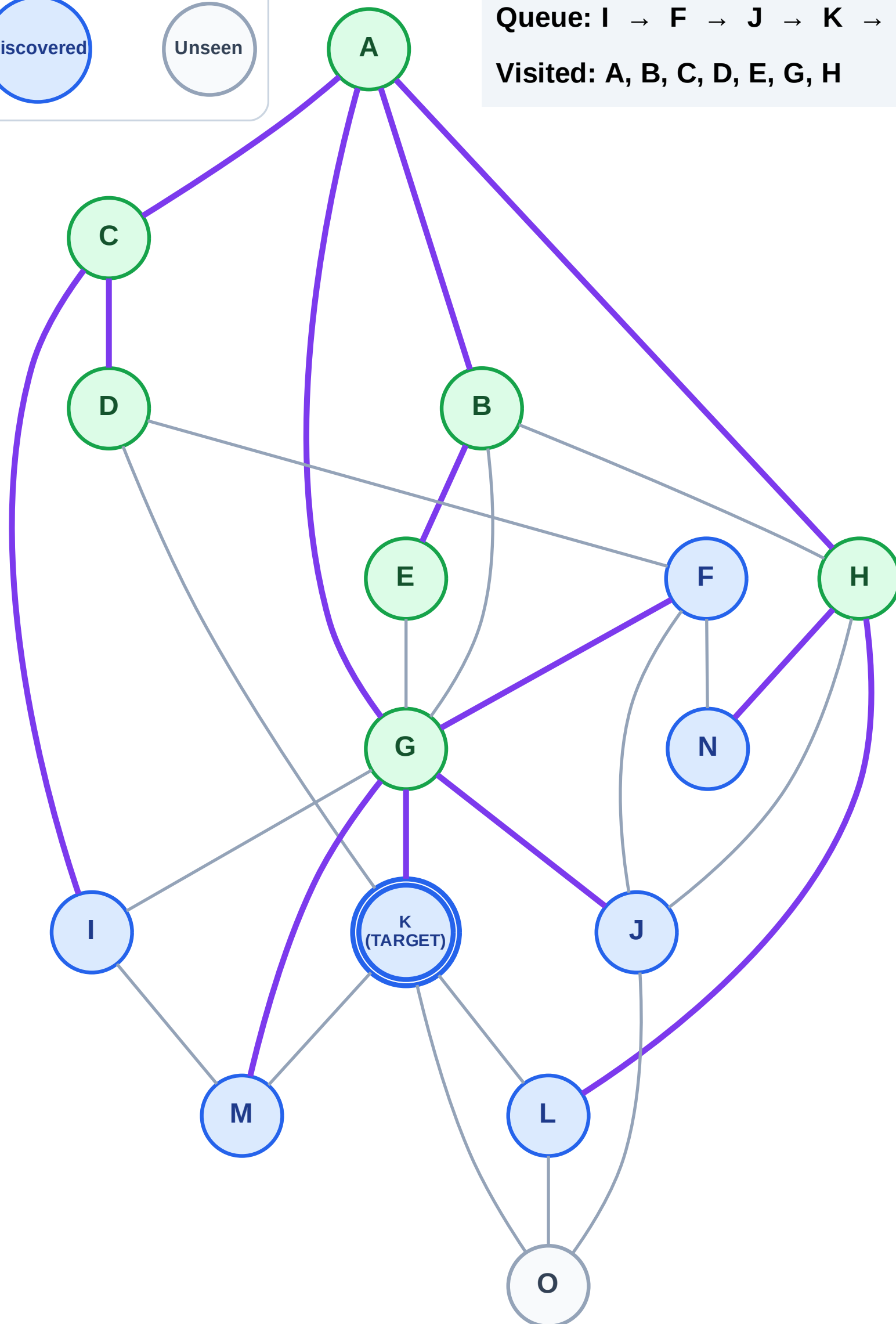
Step 15: No new neighbors from D

## Legend



Queue: I → F → J → K → M → L → N

Visited: A, B, C, D, E, G, H



# BFS Traversal

Step 16: Process node I

Legend

Complete

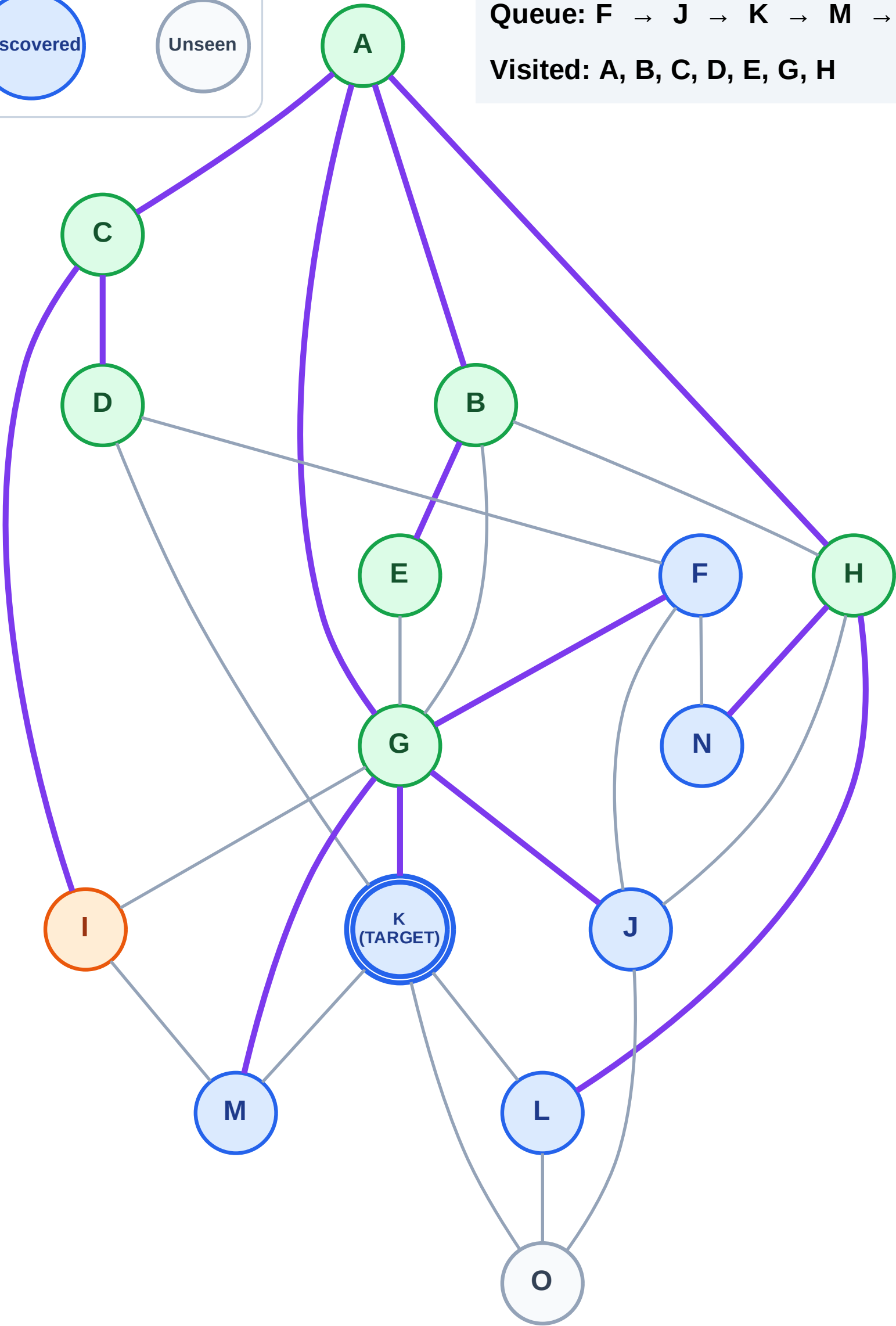
Processing

Discovered

Unseen

Queue: F → J → K → M → L → N

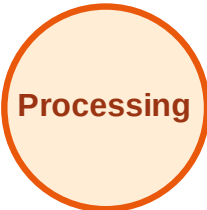
Visited: A, B, C, D, E, G, H



# BFS Traversal

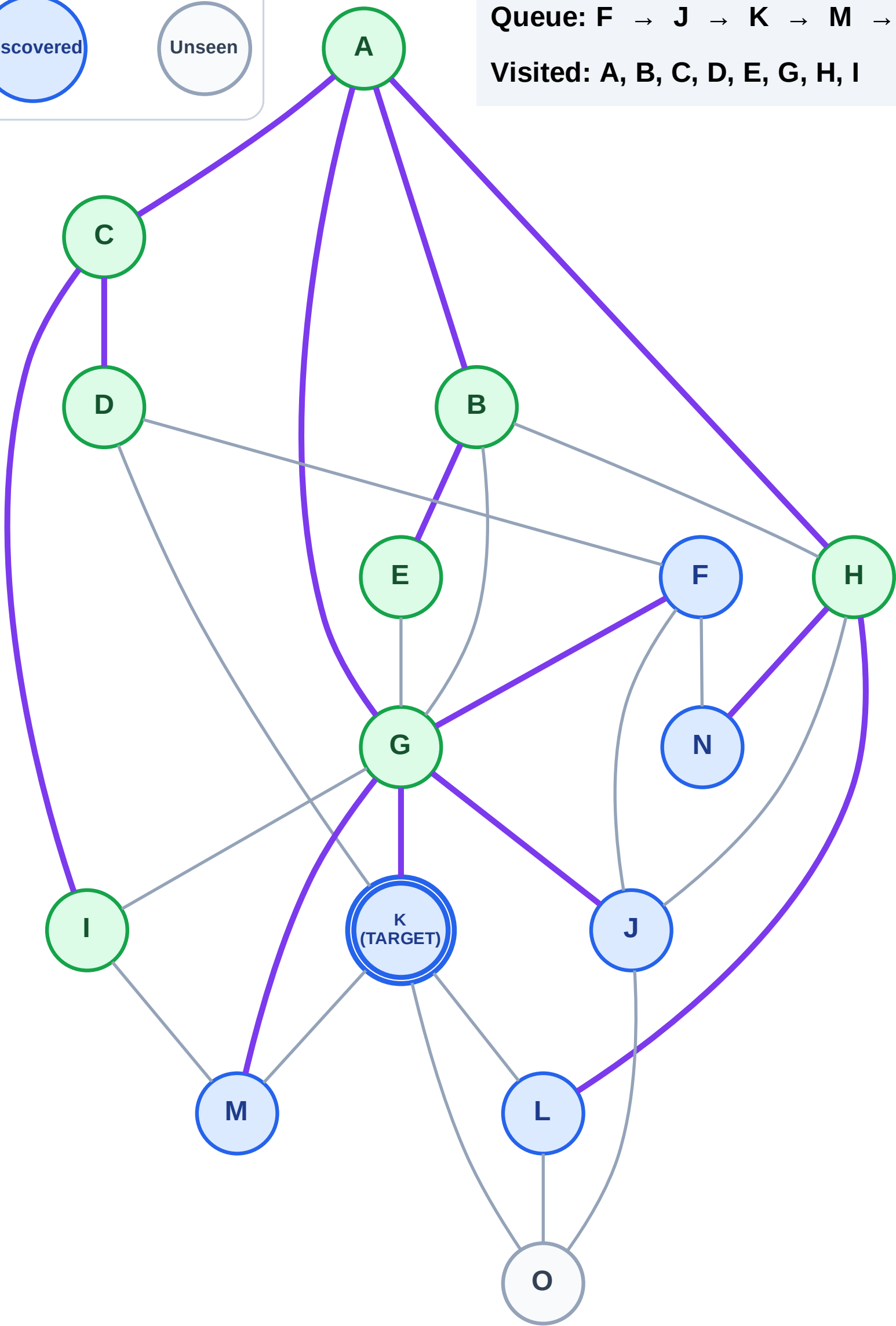
Step 17: No new neighbors from I

## Legend



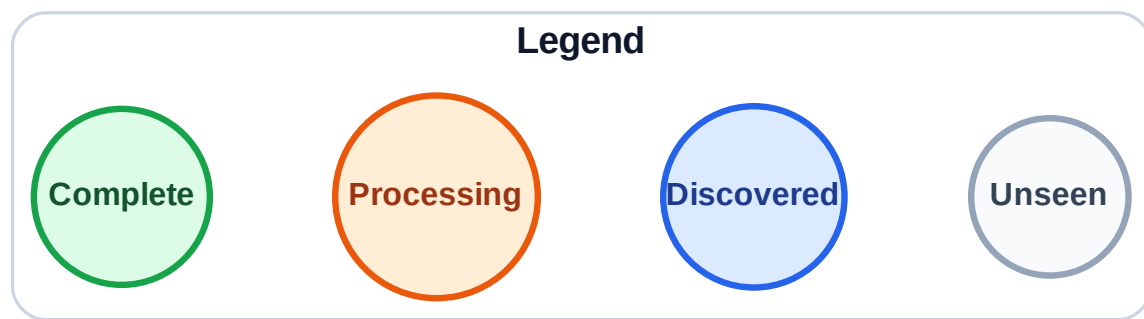
Queue: F → J → K → M → L → N

Visited: A, B, C, D, E, G, H, I



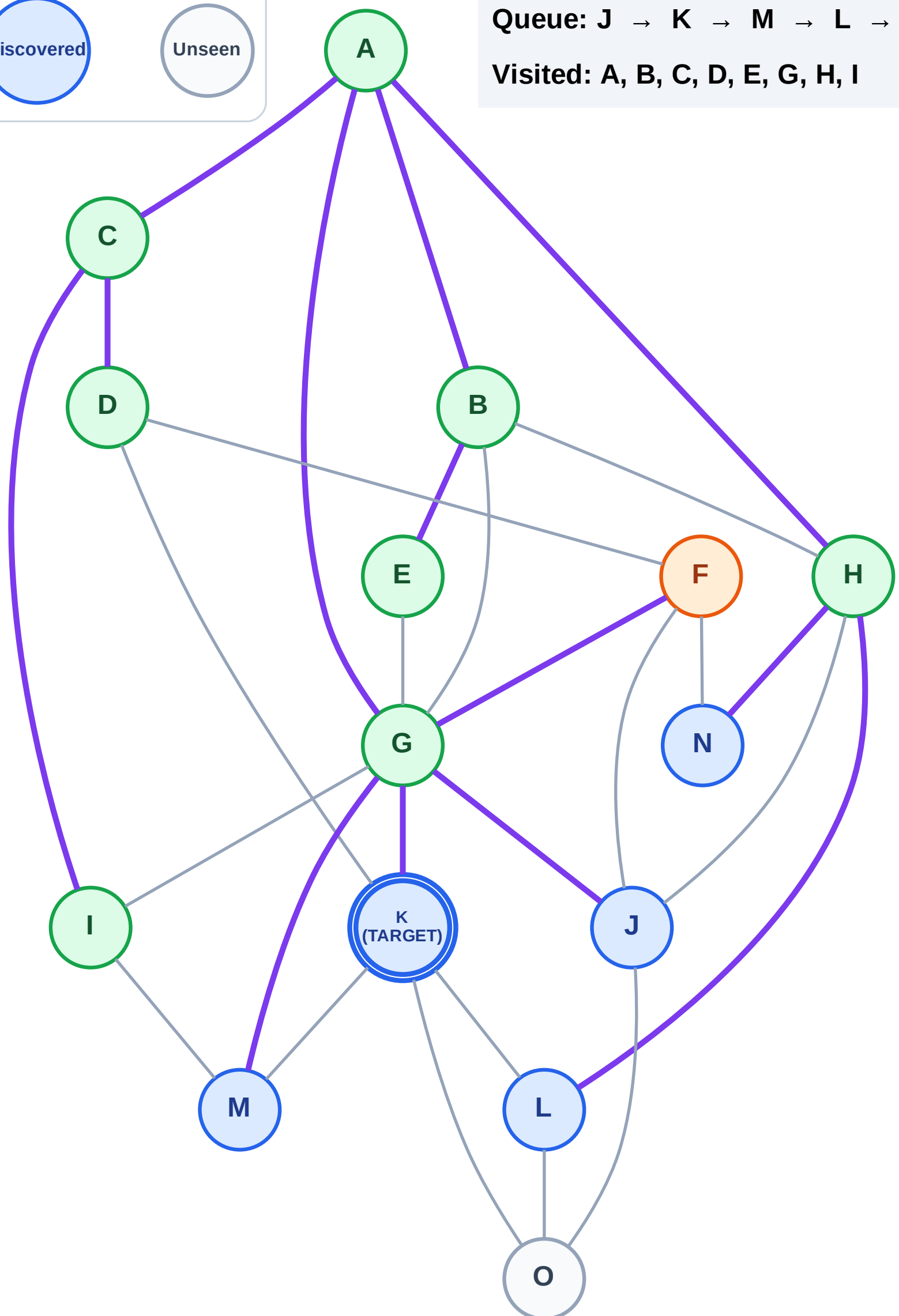
# BFS Traversal

Step 18: Process node F



Queue: J → K → M → L → N

Visited: A, B, C, D, E, G, H, I



# BFS Traversal

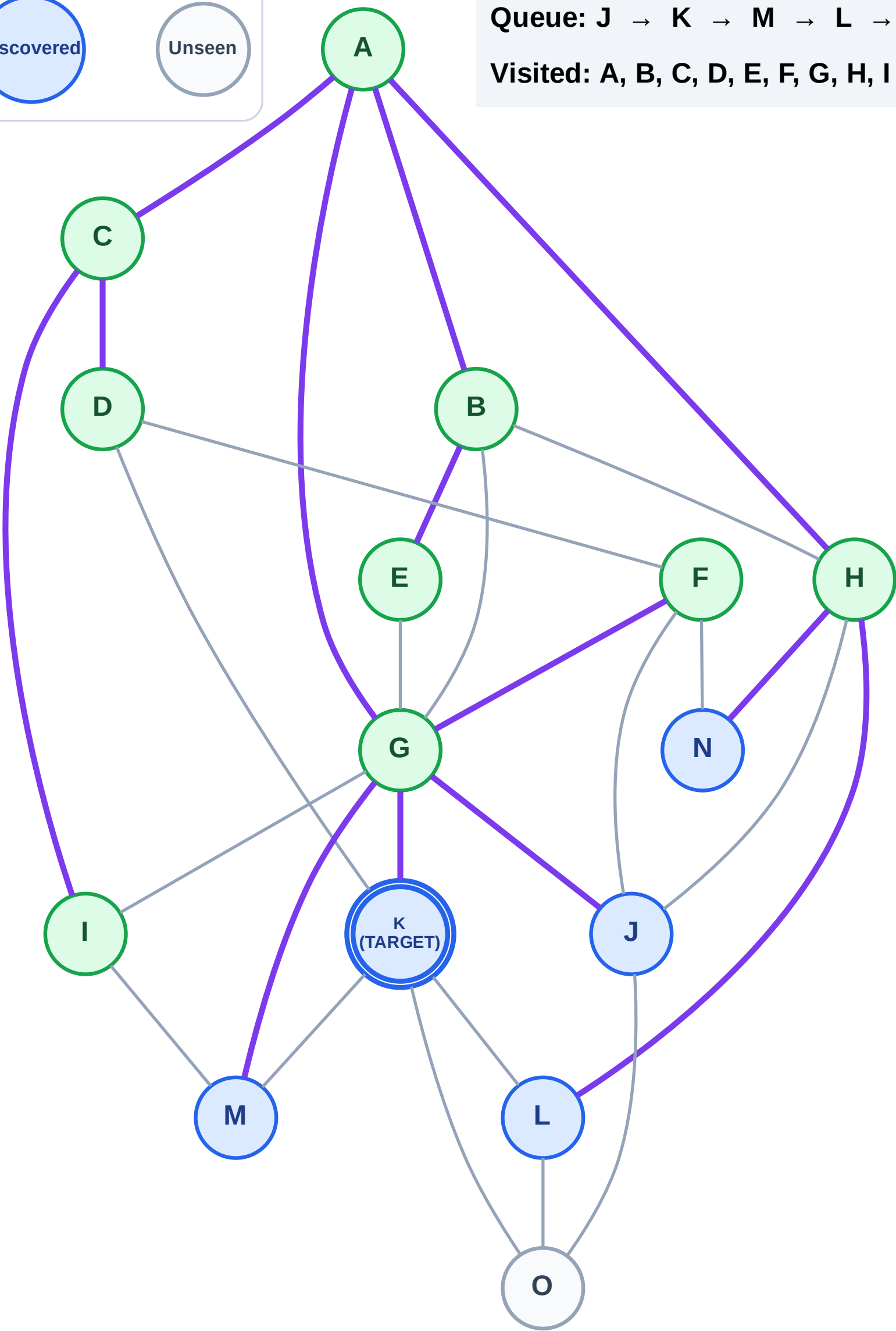
Step 19: No new neighbors from F

## Legend



Queue: J → K → M → L → N

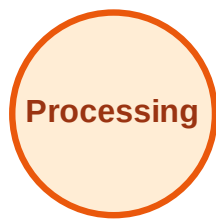
Visited: A, B, C, D, E, F, G, H, I



# BFS Traversal

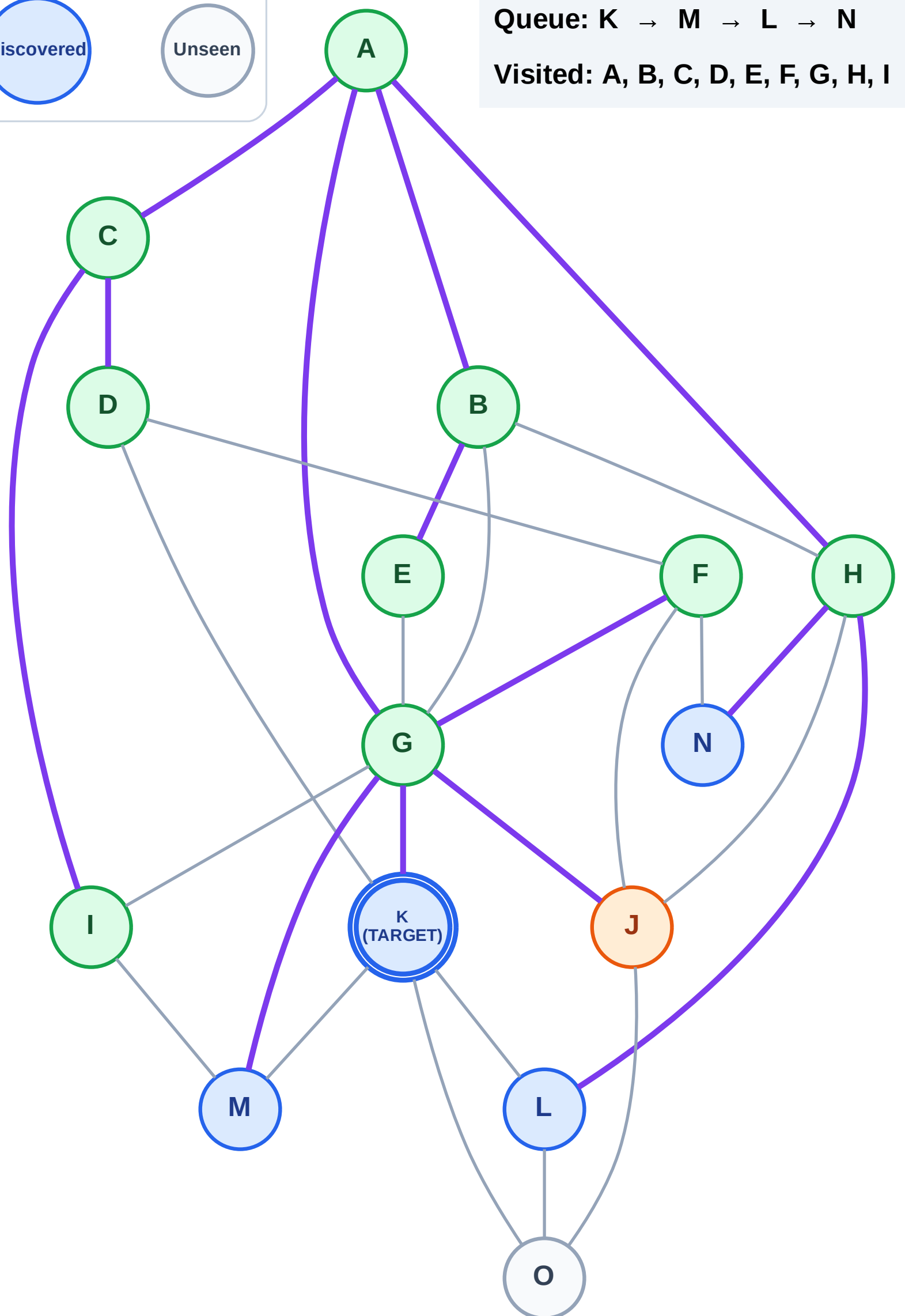
Step 20: Process node J

## Legend



Queue: K → M → L → N

Visited: A, B, C, D, E, F, G, H, I



# BFS Traversal

Step 21: Discover O from J

Legend

Complete

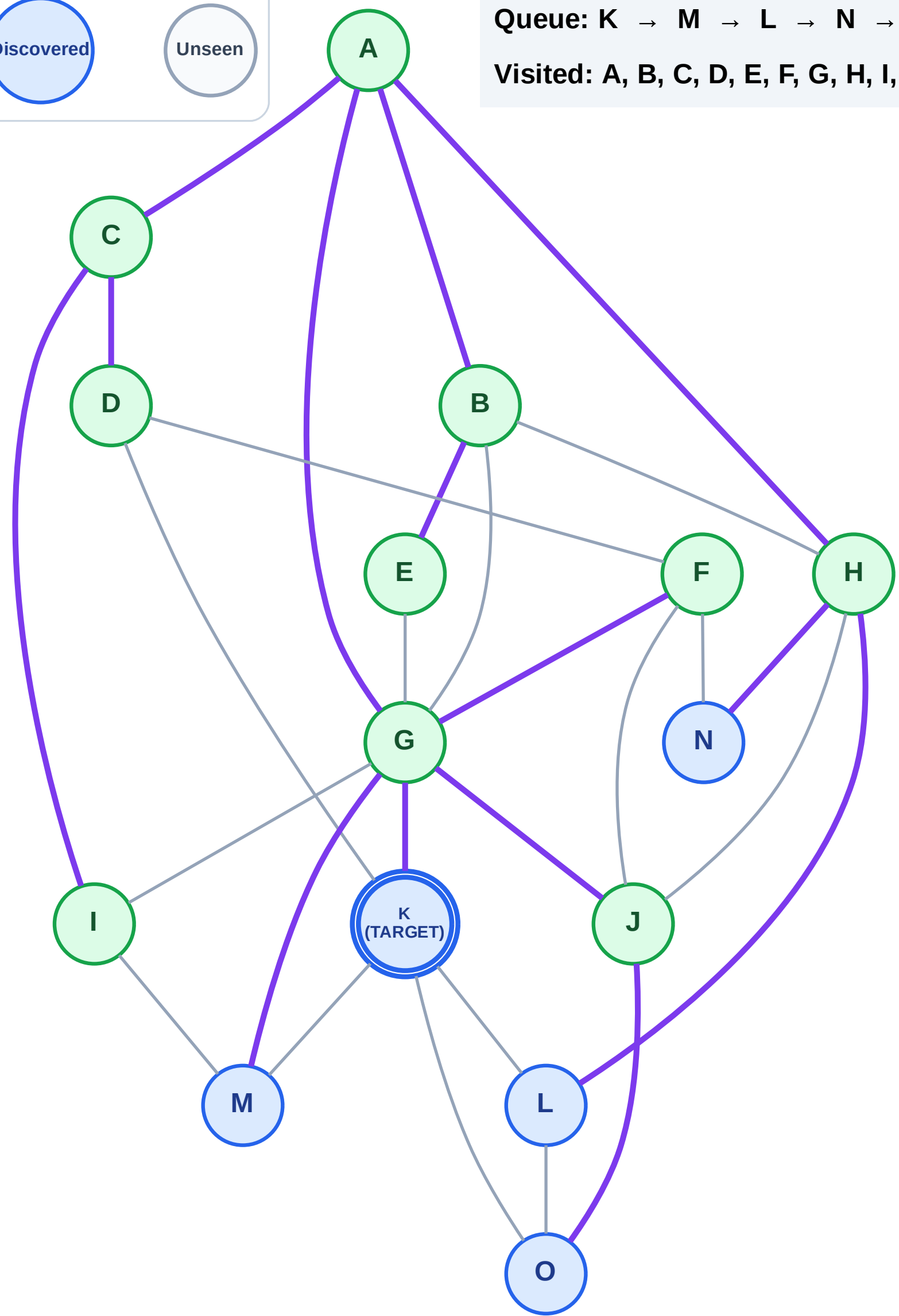
Processing

Discovered

Unseen

Queue: K → M → L → N → O

Visited: A, B, C, D, E, F, G, H, I, J



# BFS Traversal

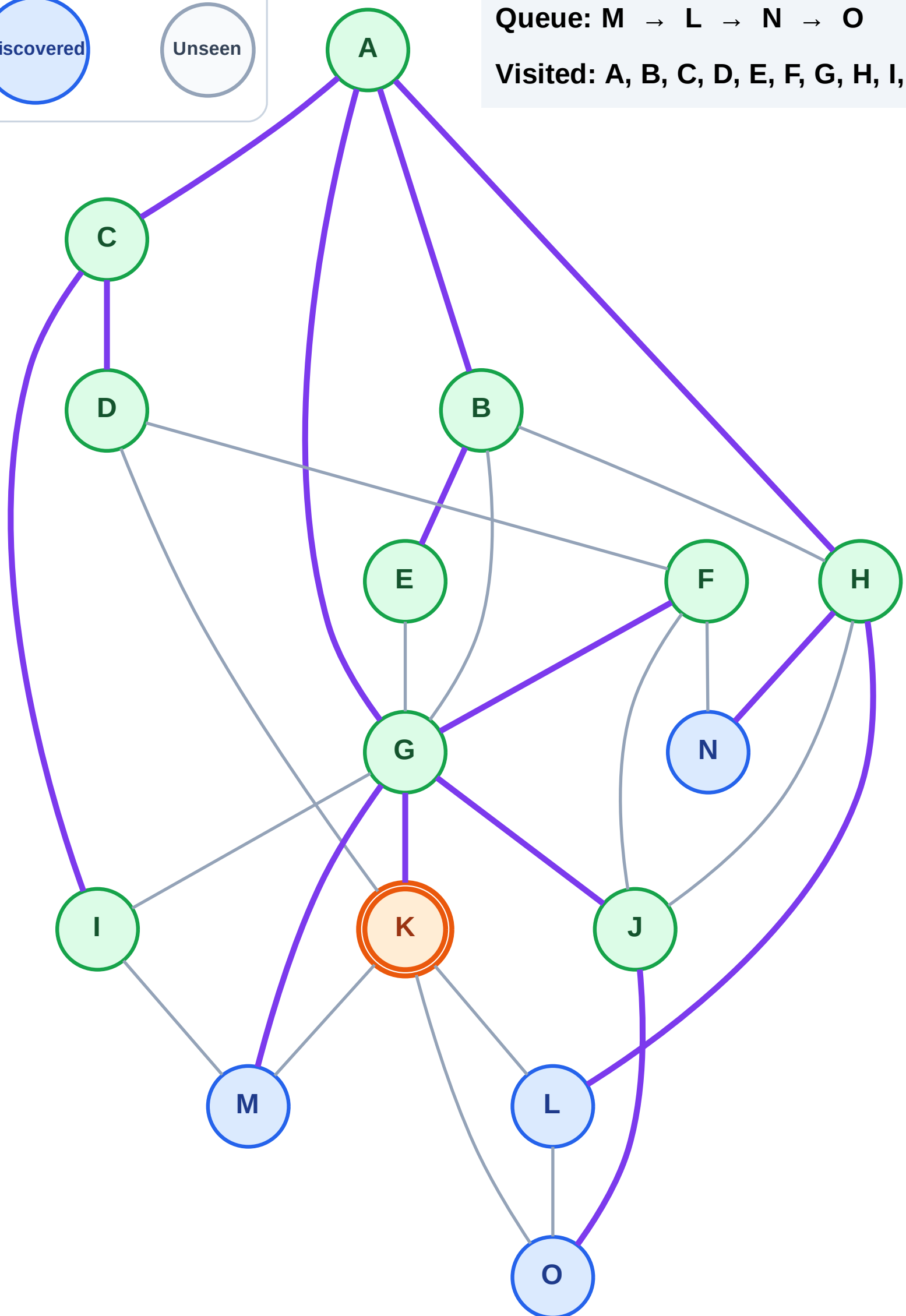
Step 22: Process node K

## Legend



Queue: M → L → N → O

Visited: A, B, C, D, E, F, G, H, I, J



# BFS Traversal

Step 23: TARGET NODE K REACHED! Path (2 hops): A → G → K

Legend

Complete

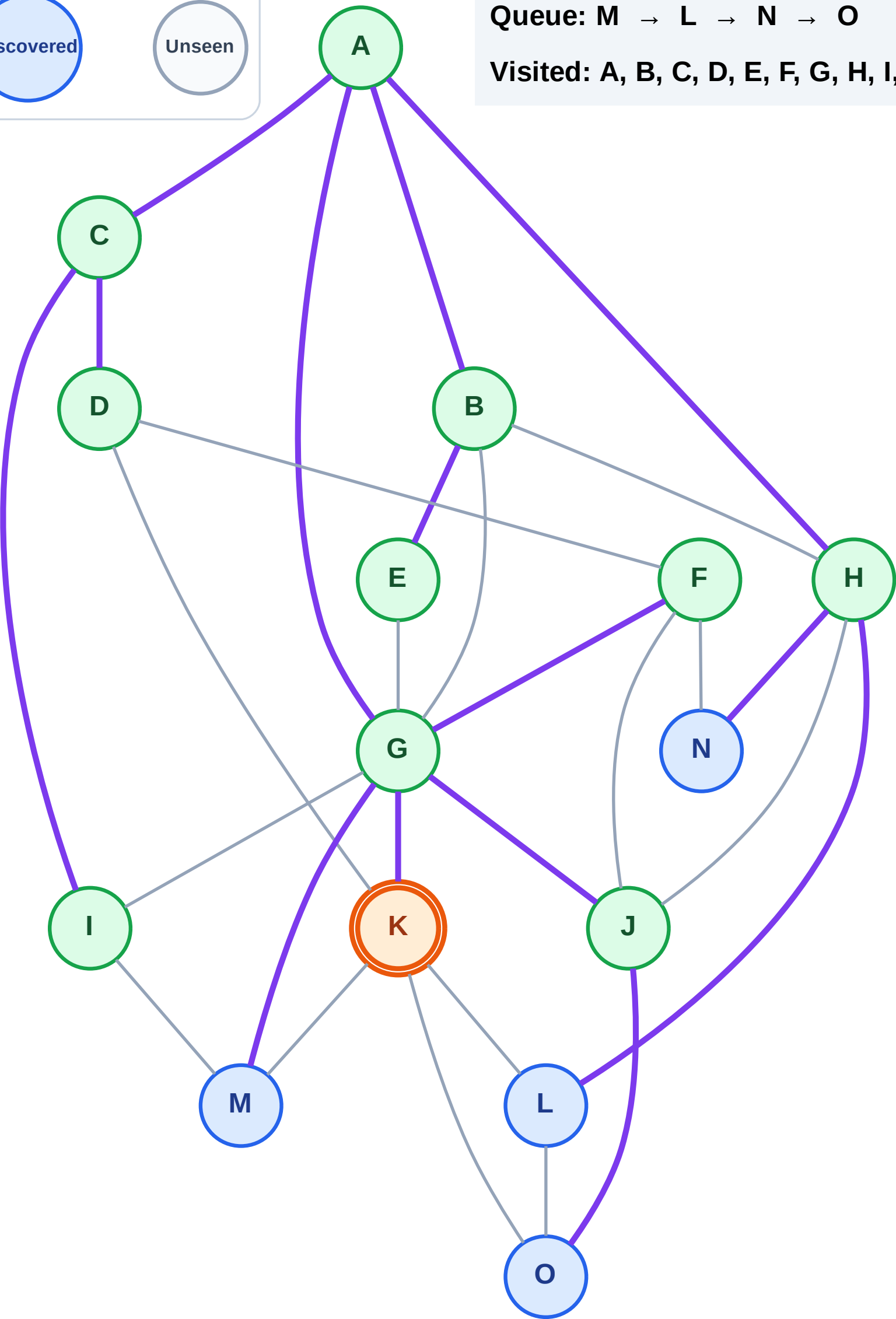
Processing

Discovered

Unseen

Queue: M → L → N → O

Visited: A, B, C, D, E, F, G, H, I, J, K



# DFS Traversal

Step 1: Start at A (Seeking target node K)

## Legend

Complete

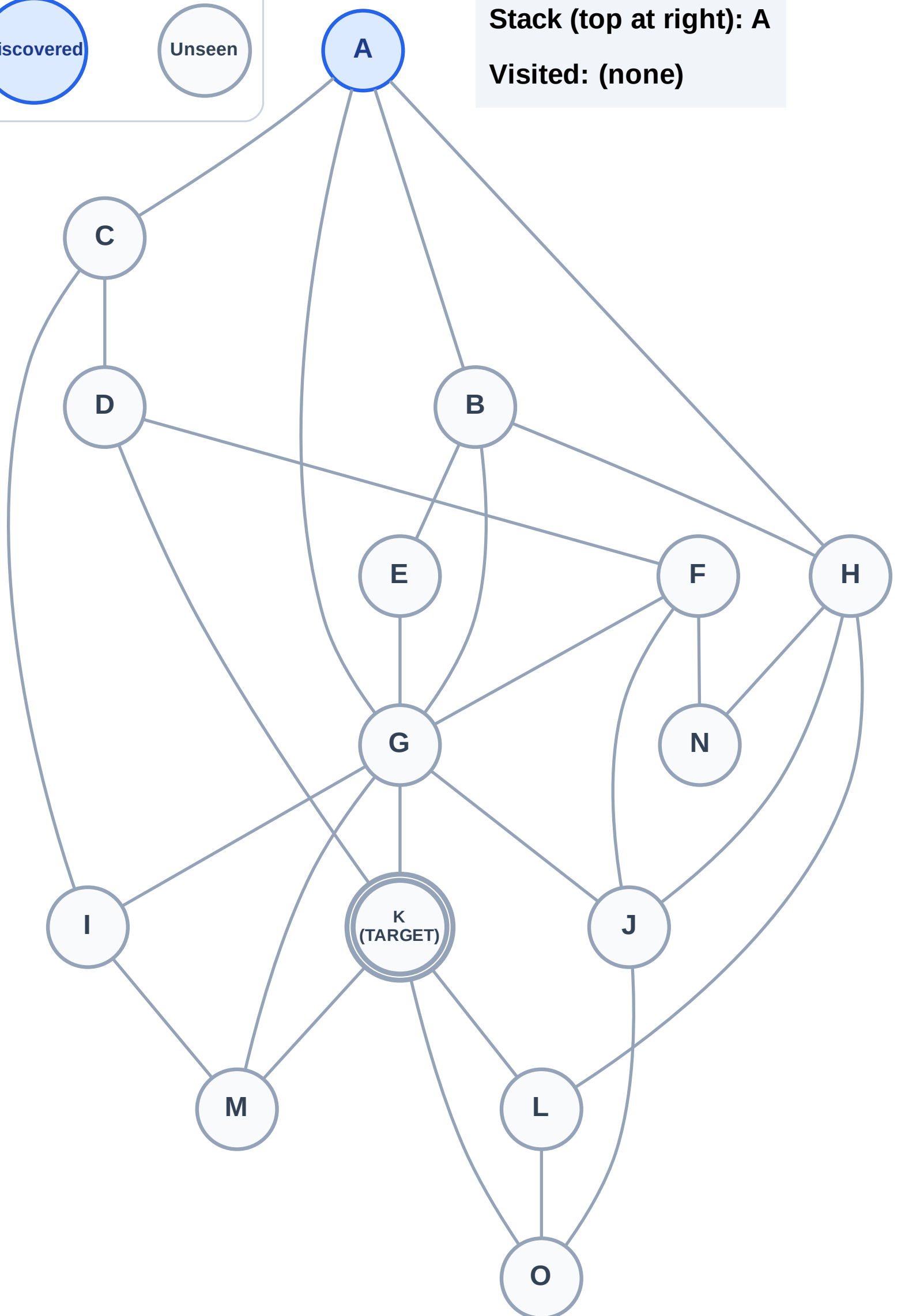
Processing

Discovered

Unseen

Stack (top at right): A

Visited: (none)



# DFS Traversal

Step 2: Process node A

Legend

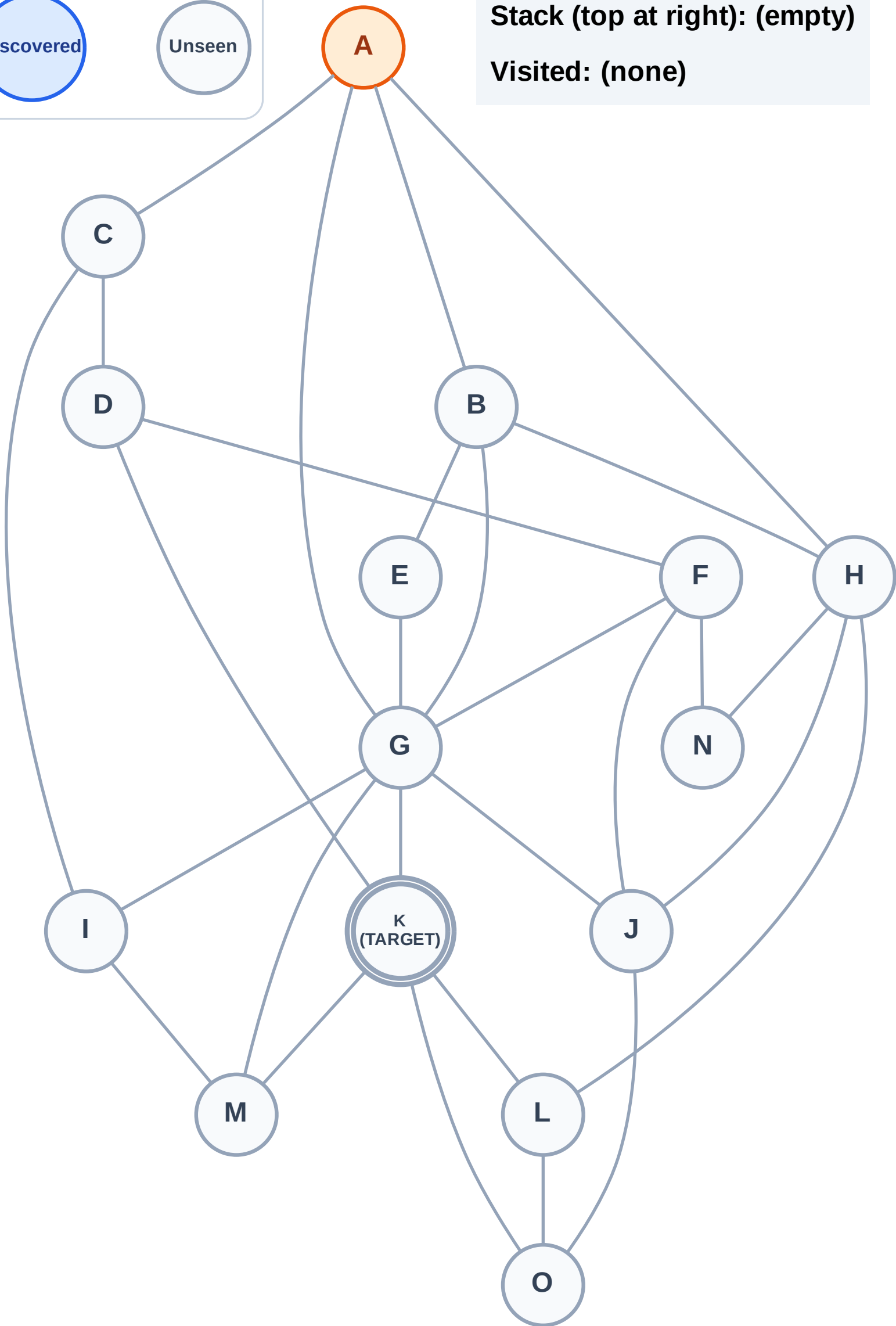
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)  
Visited: (none)



# DFS Traversal

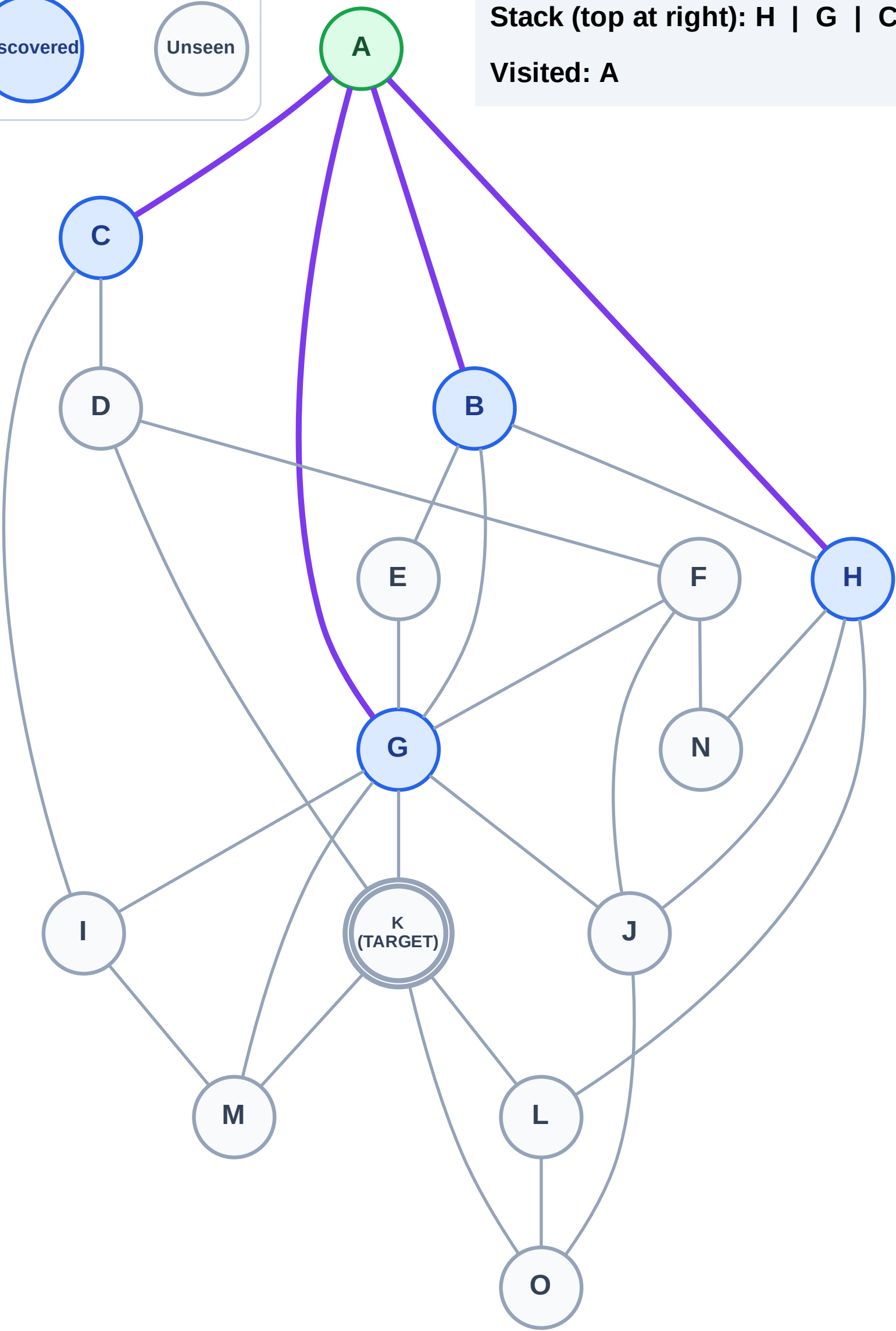
Step 3: Discover H, G, C, B from A

## Legend



Stack (top at right): H | G | C | B

Visited: A



# DFS Traversal

Step 4: Process node B

Legend

Complete

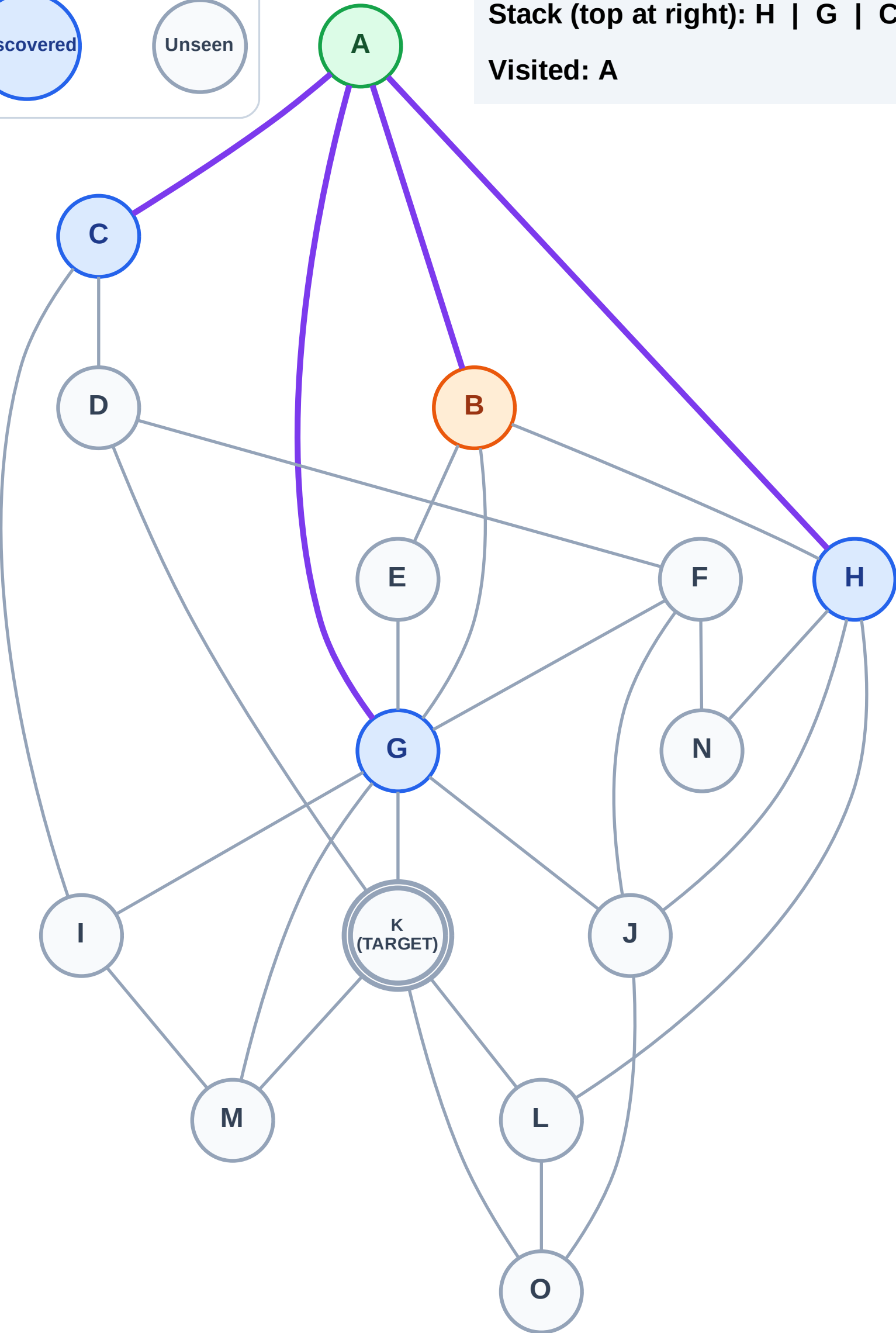
Processing

Discovered

Unseen

Stack (top at right): H | G | C

Visited: A





# DFS Traversal

Step 6: Process node E

Legend

Complete

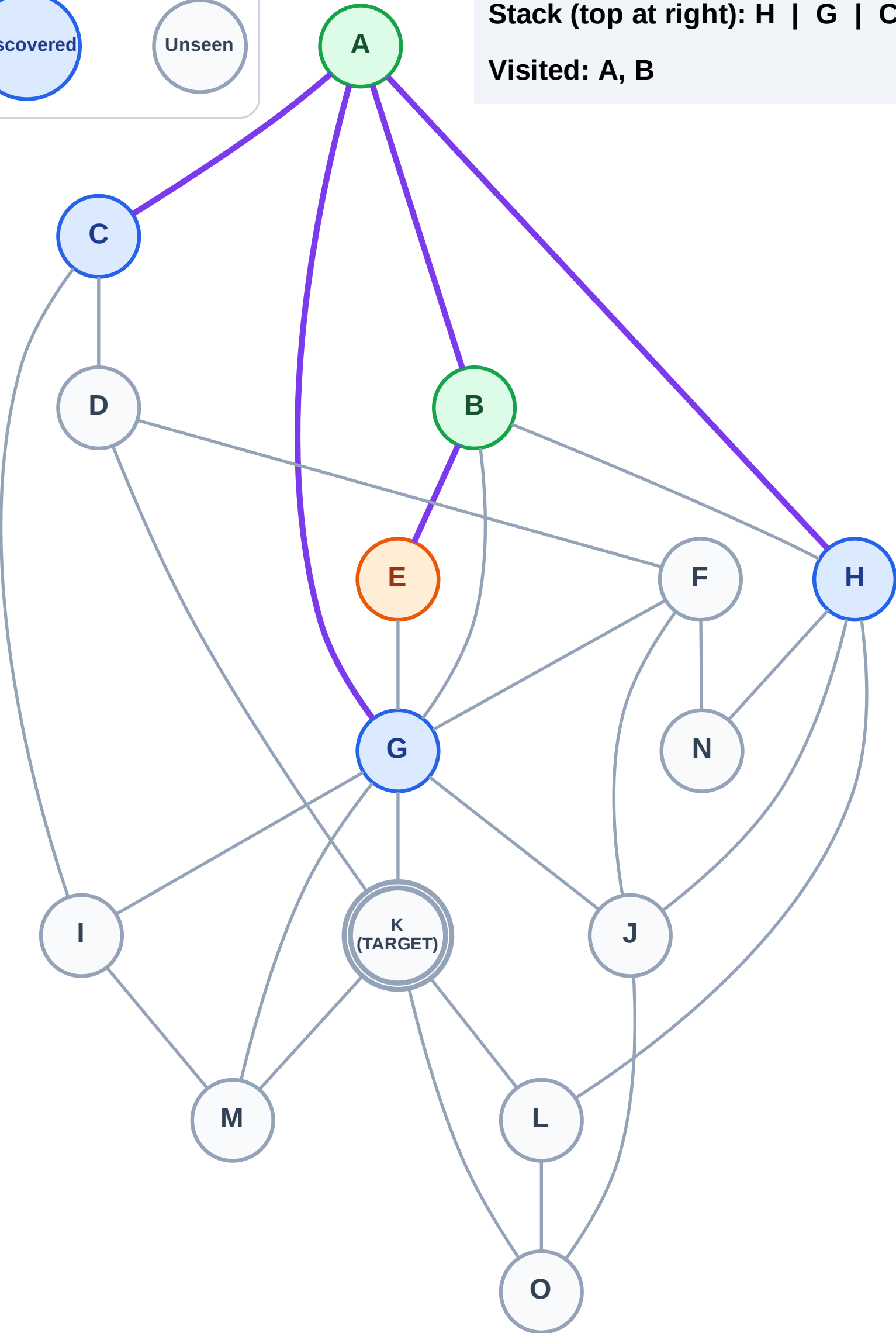
Processing

Discovered

Unseen

Stack (top at right): H | G | C

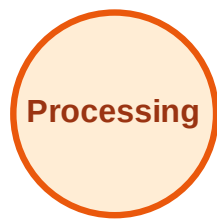
Visited: A, B



# DFS Traversal

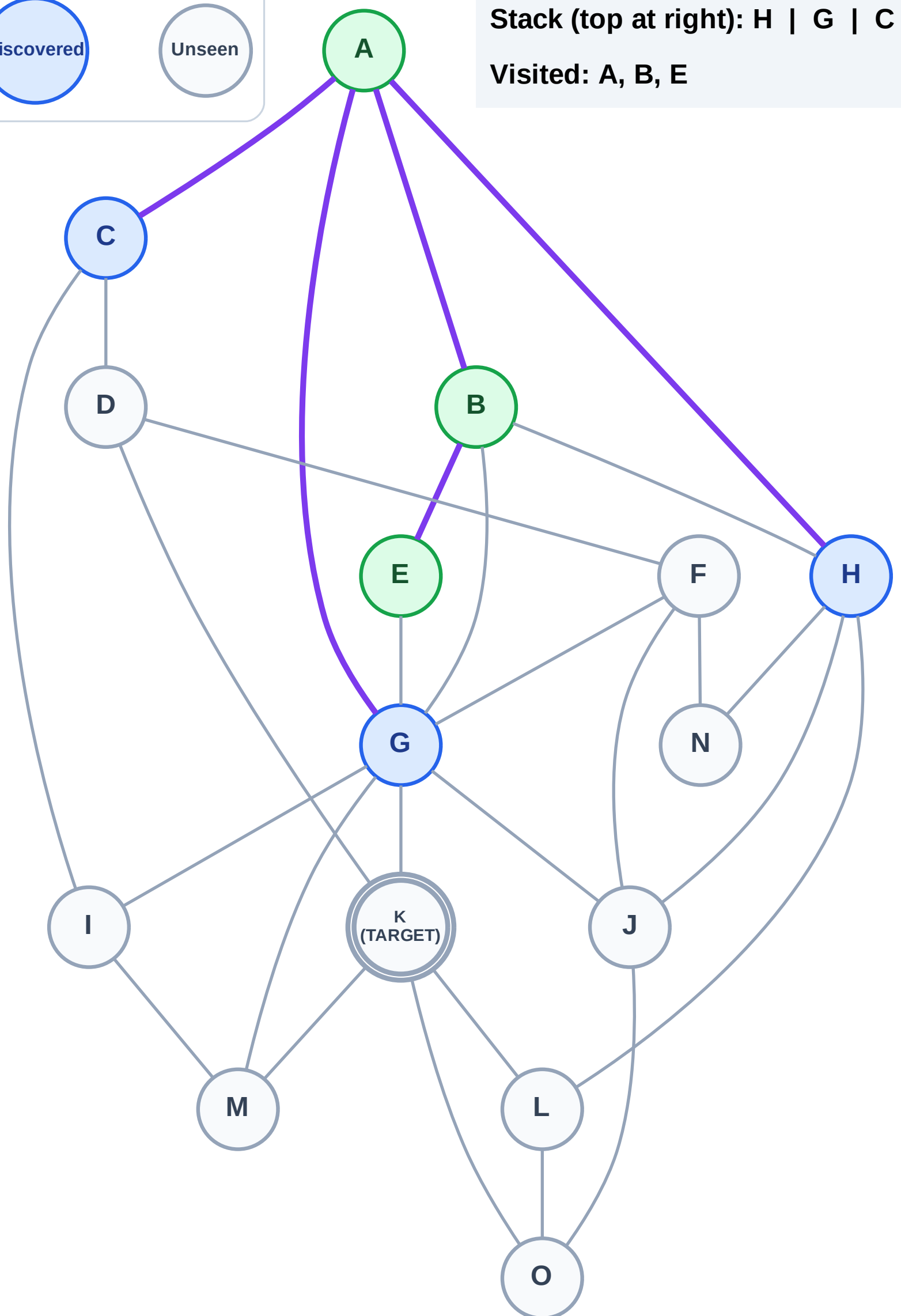
Step 7: No new neighbors from E

## Legend



Stack (top at right): H | G | C

Visited: A, B, E



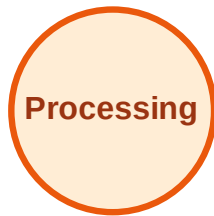
# DFS Traversal

Step 8: Process node C

## Legend



Complete



Processing



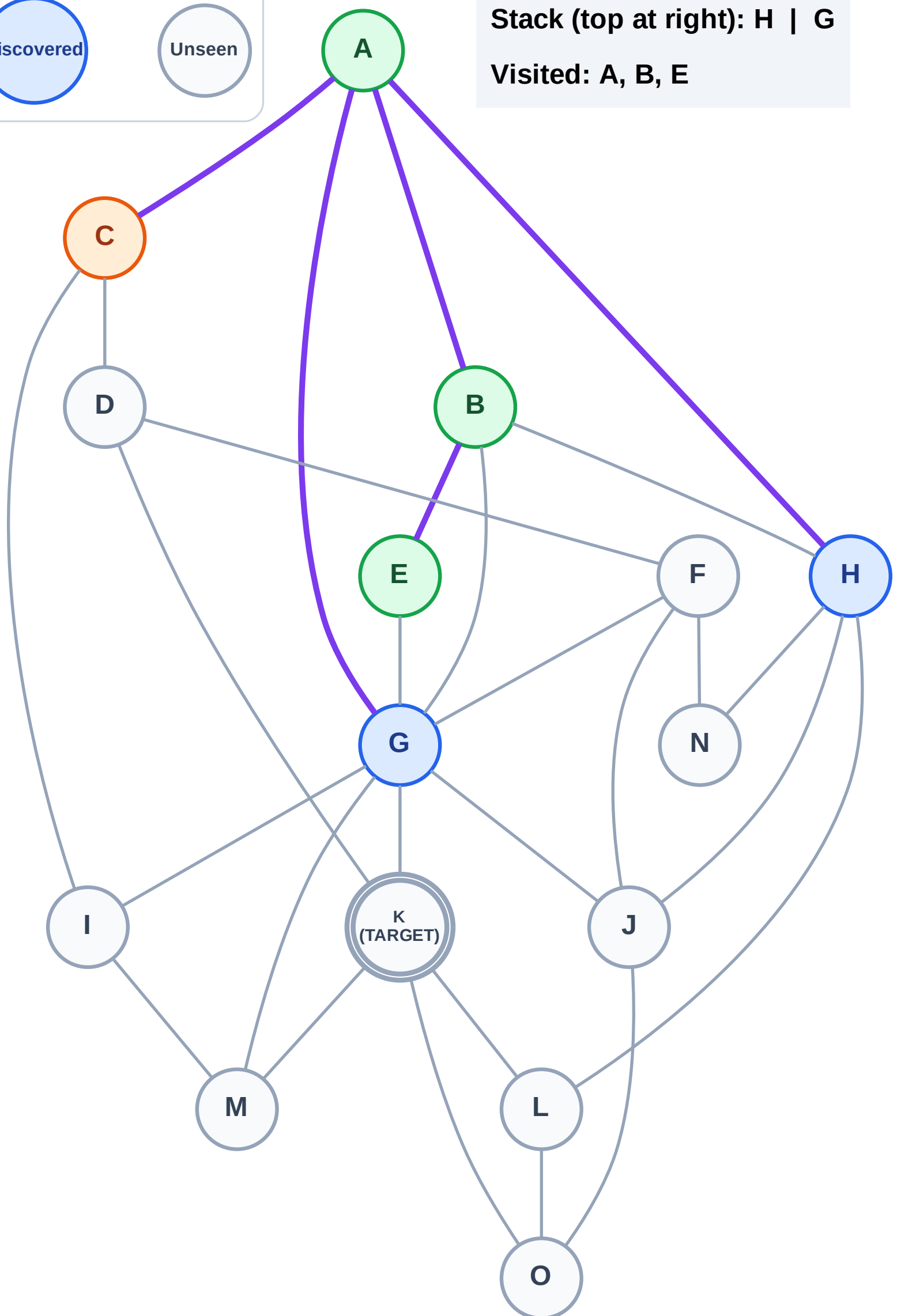
Discovered



Unseen

Stack (top at right): H | G

Visited: A, B, E



# DFS Traversal

Step 9: Discover I, D from C

## Legend



Complete



Processing



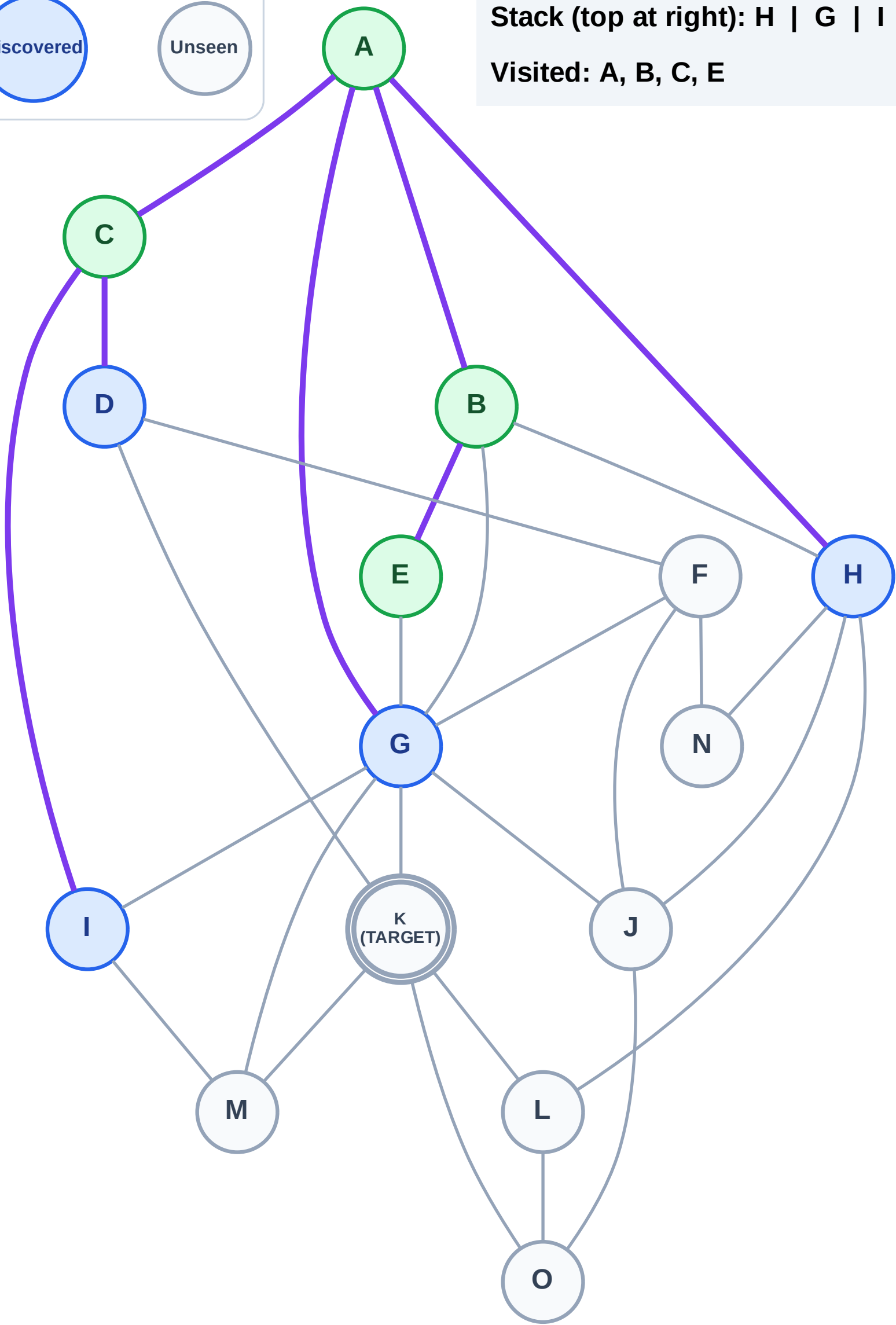
Discovered



Unseen

Stack (top at right): H | G | I | D

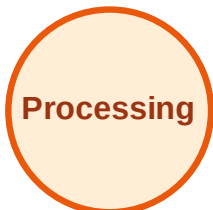
Visited: A, B, C, E



# DFS Traversal

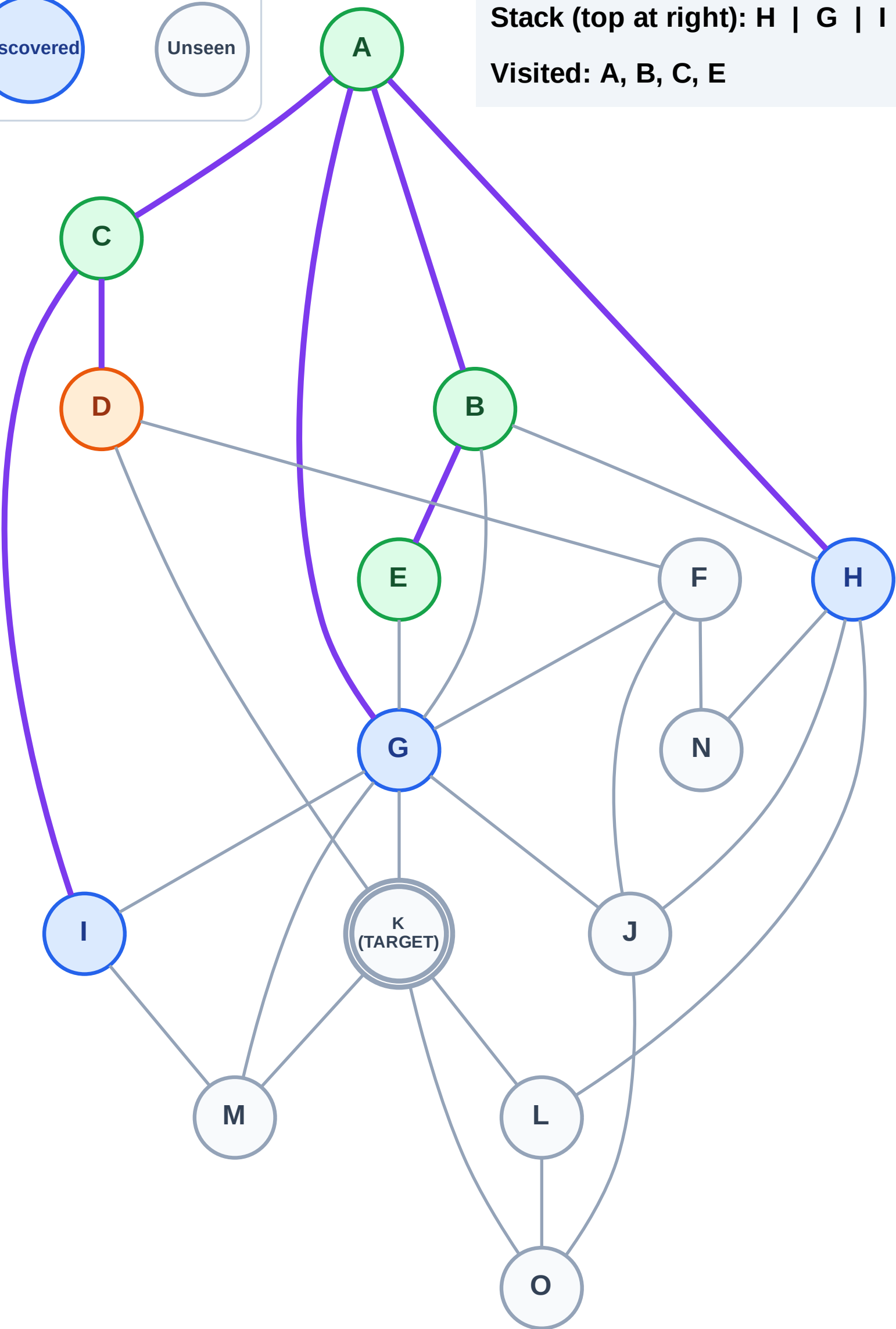
Step 10: Process node D

## Legend



Stack (top at right): H | G | I

Visited: A, B, C, E



# DFS Traversal

Step 11: Discover K, F from D

## Legend



Complete



Processing



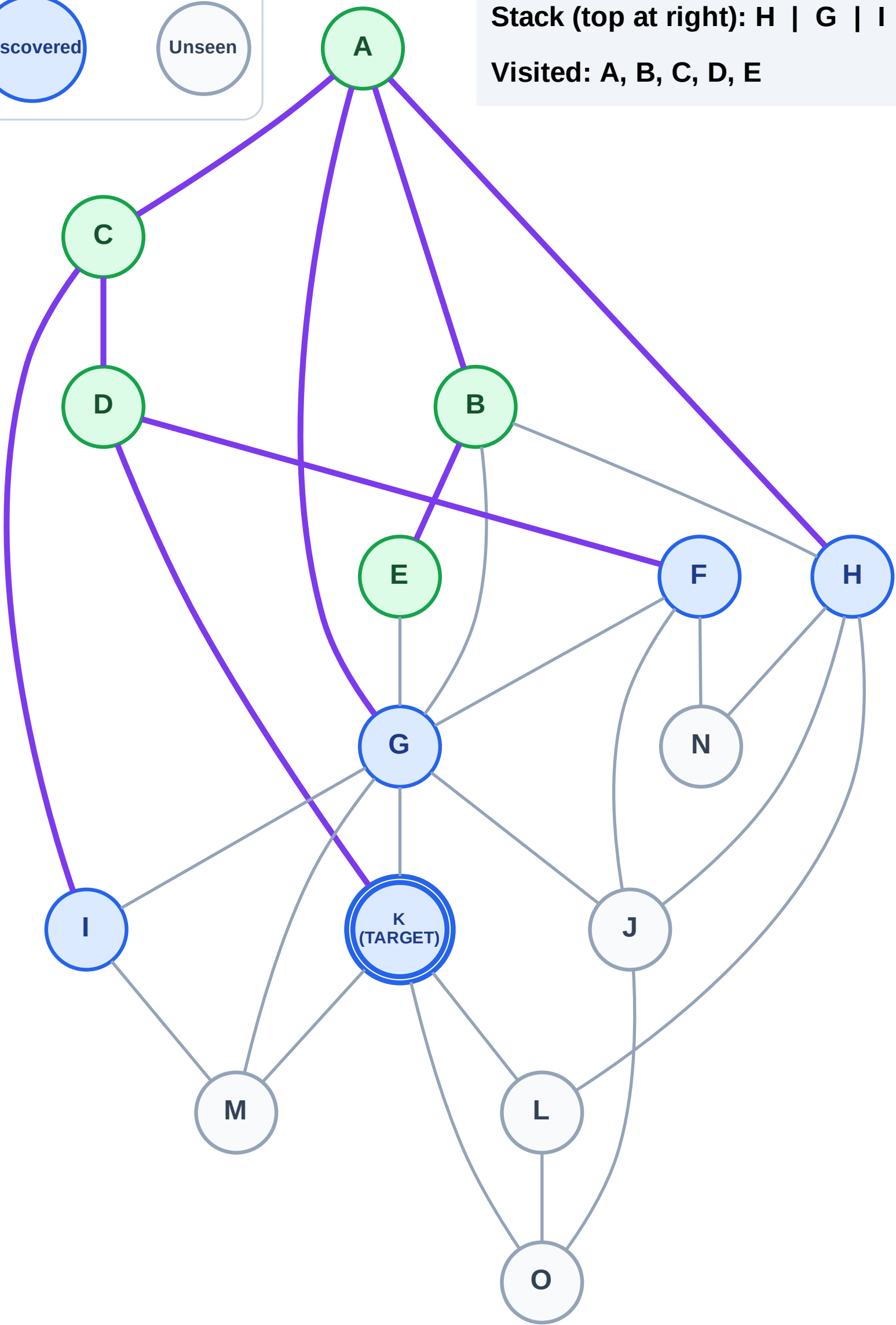
Discovered



Unseen

Stack (top at right): H | G | I | K | F

Visited: A, B, C, D, E



# DFS Traversal

Step 12: Process node F

Legend

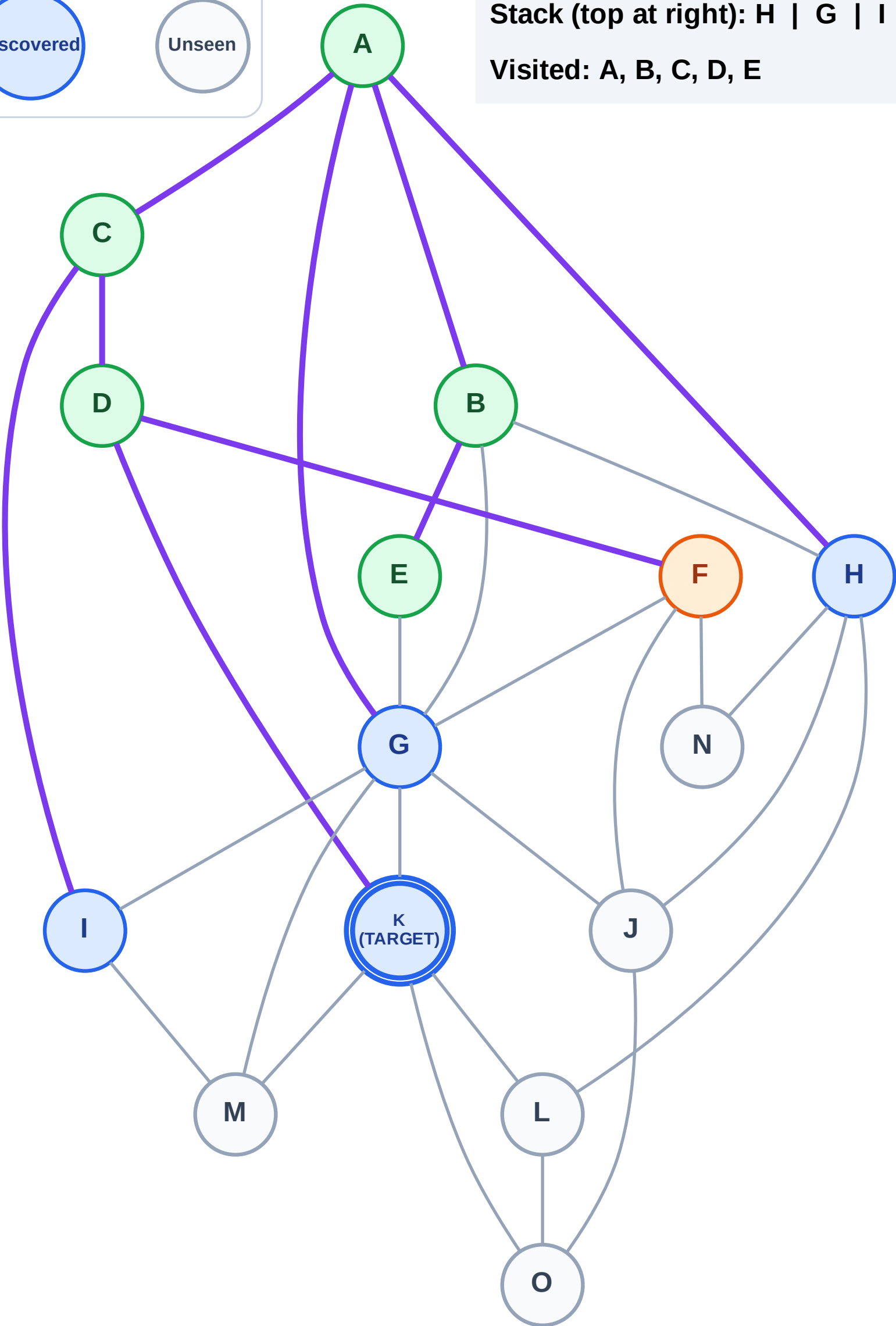
Complete

Processing

Discovered

Unseen

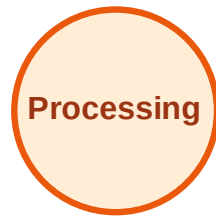
Stack (top at right): H | G | I | K  
Visited: A, B, C, D, E



# DFS Traversal

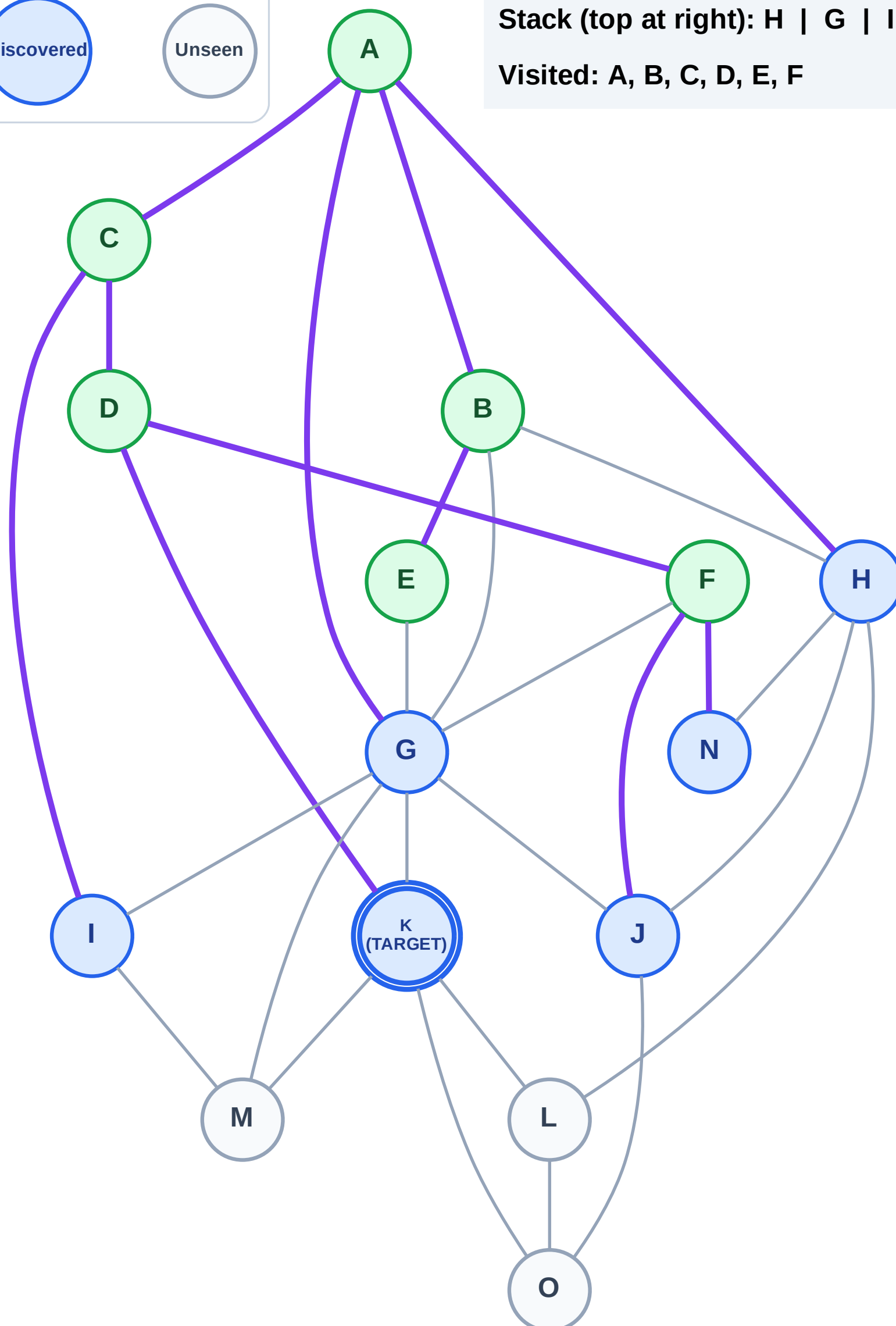
Step 13: Discover N, J from F

## Legend



Stack (top at right): H | G | I | K | N | J

Visited: A, B, C, D, E, F



# DFS Traversal

Step 14: Process node J

Legend

Complete

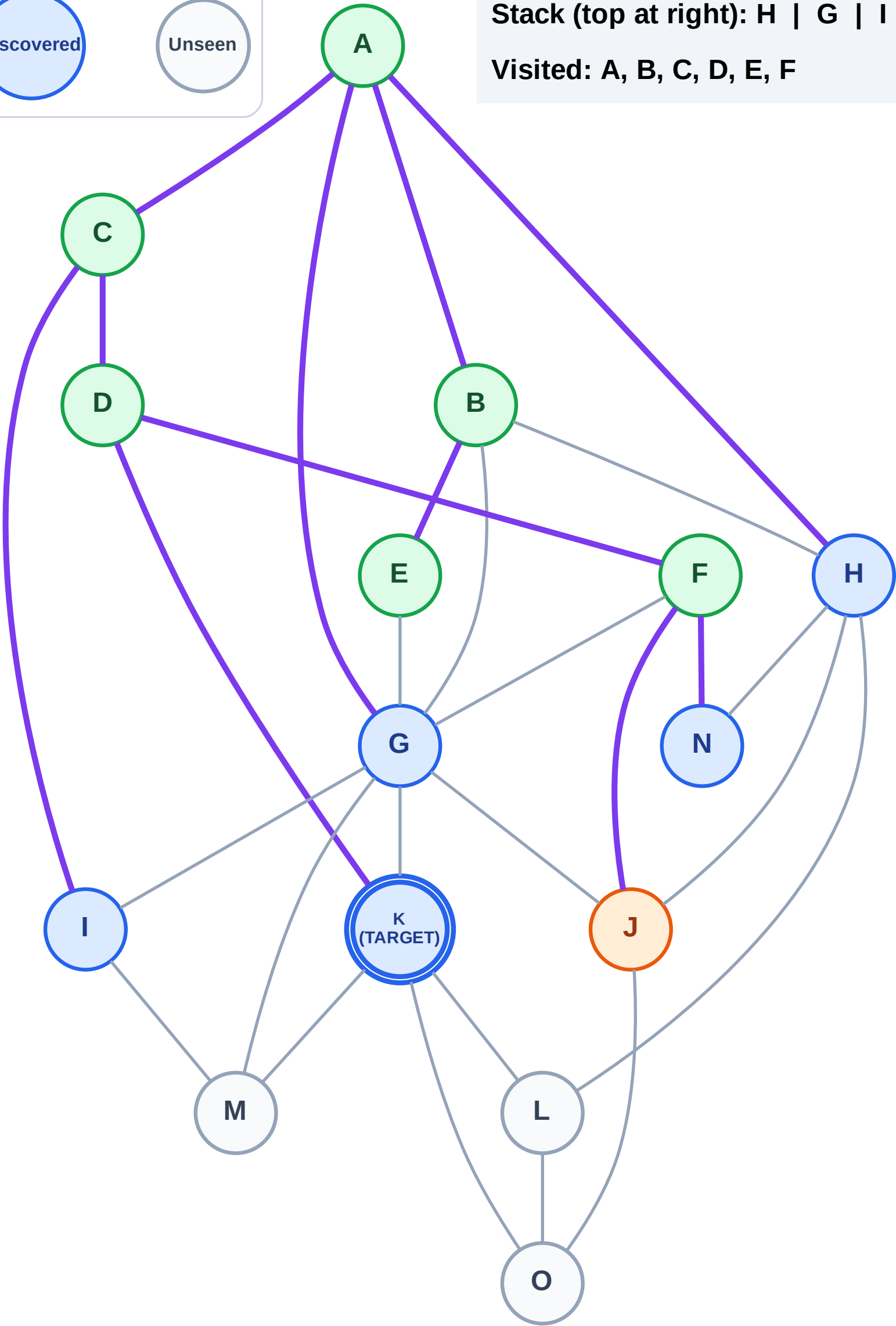
Processing

Discovered

Unseen

Stack (top at right): H | G | I | K | N

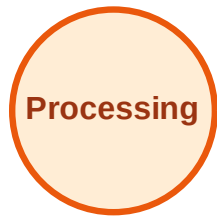
Visited: A, B, C, D, E, F



# DFS Traversal

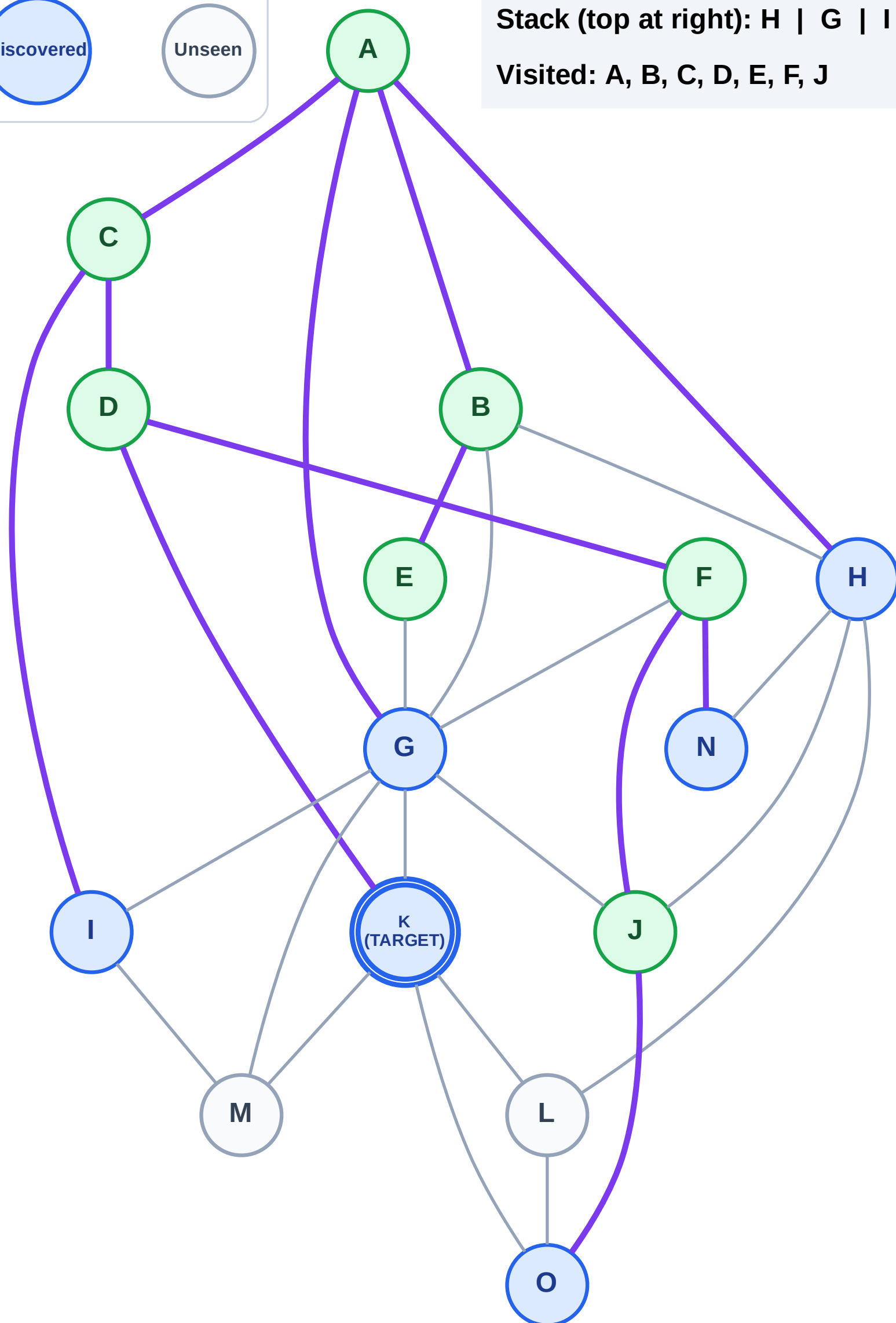
Step 15: Discover O from J

## Legend



Stack (top at right): H | G | I | K | N | O

Visited: A, B, C, D, E, F, J



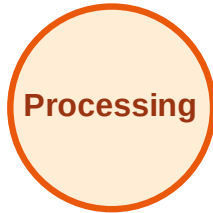
# DFS Traversal

Step 16: Process node O

## Legend



Complete



Processing



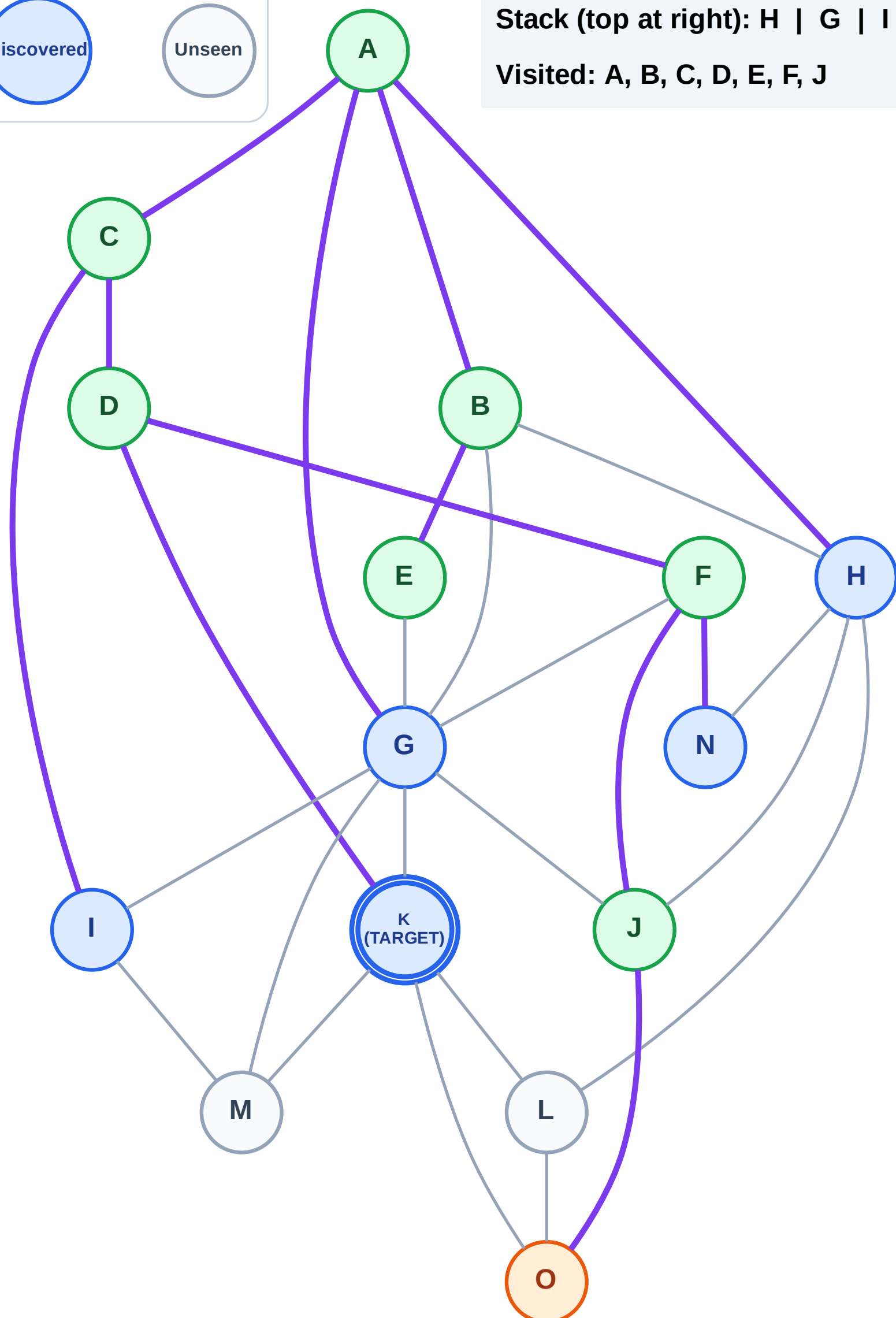
Discovered



Unseen

Stack (top at right): H | G | I | K | N

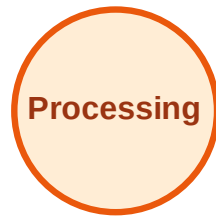
Visited: A, B, C, D, E, F, J



# DFS Traversal

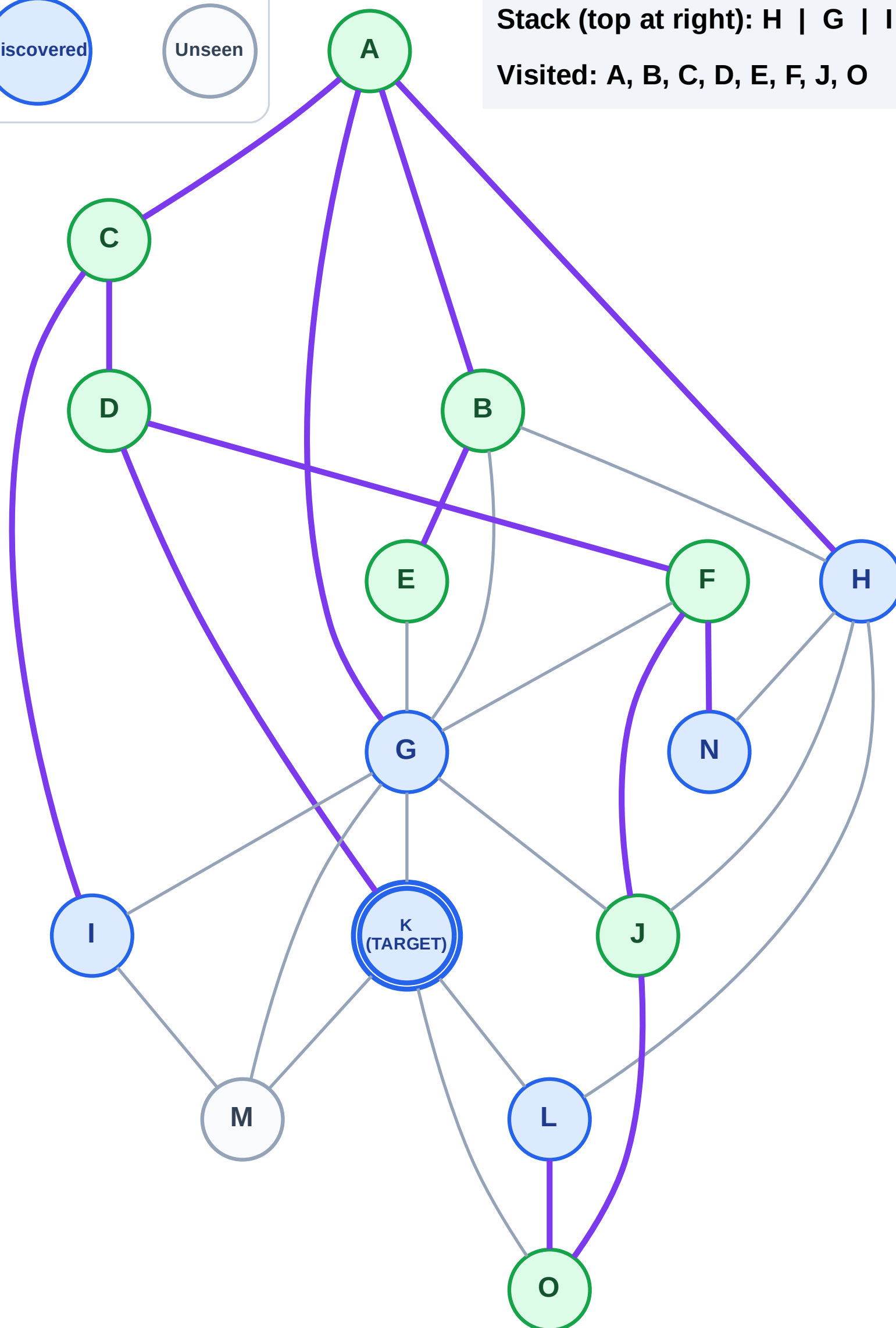
Step 17: Discover L from O

## Legend



Stack (top at right): H | G | I | K | N | L

Visited: A, B, C, D, E, F, J, O



# DFS Traversal

Step 18: Process node L

Stack (top at right): H | G | I | K | N  
Visited: A, B, C, D, E, F, J, O

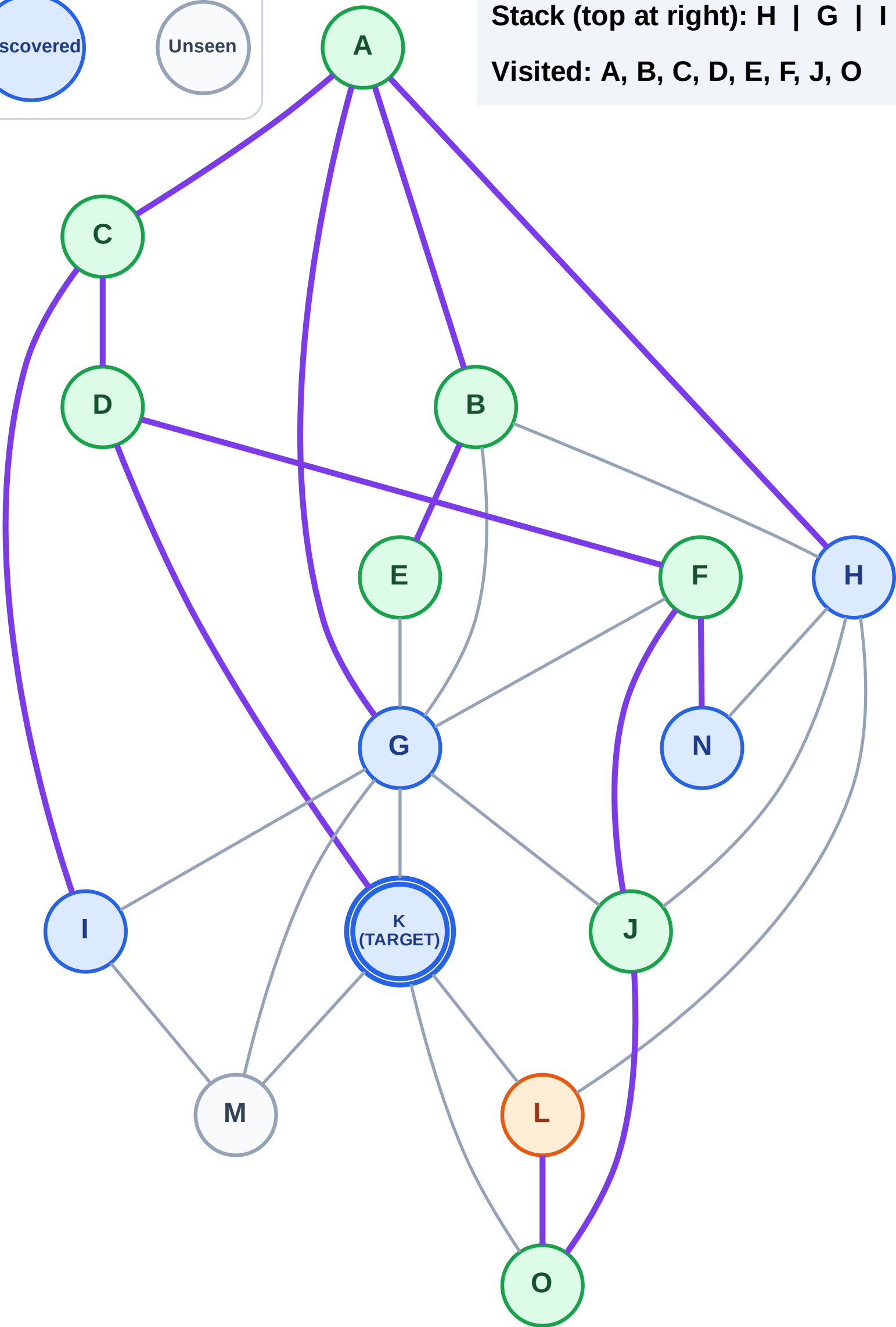
Legend

Complete

Processing

Discovered

Unseen



# DFS Traversal

Step 19: No new neighbors from L

## Legend



Complete



Processing



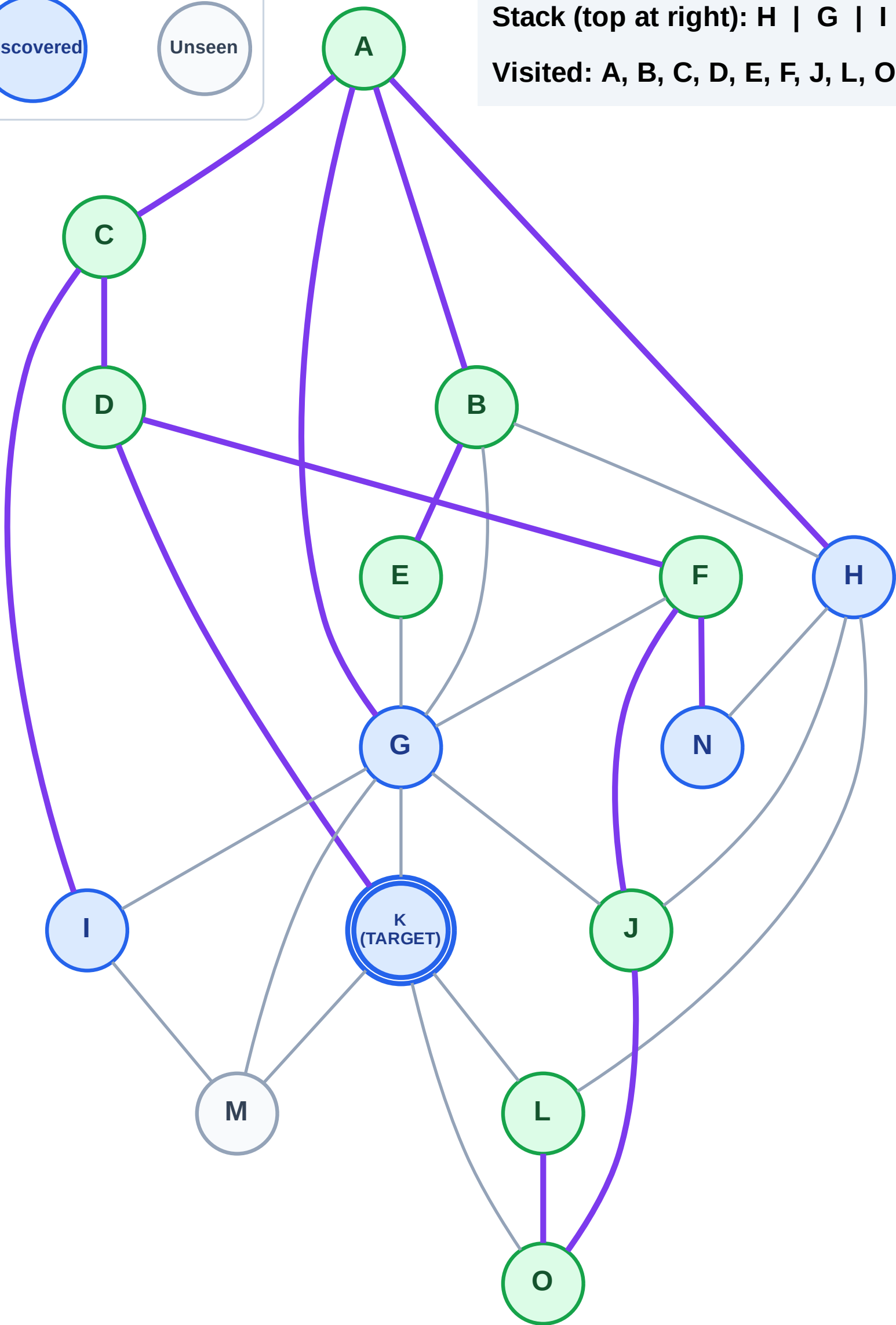
Discovered



Unseen

Stack (top at right): H | G | I | K | N

Visited: A, B, C, D, E, F, J, L, O



# DFS Traversal

Step 20: Process node N

Legend

Complete

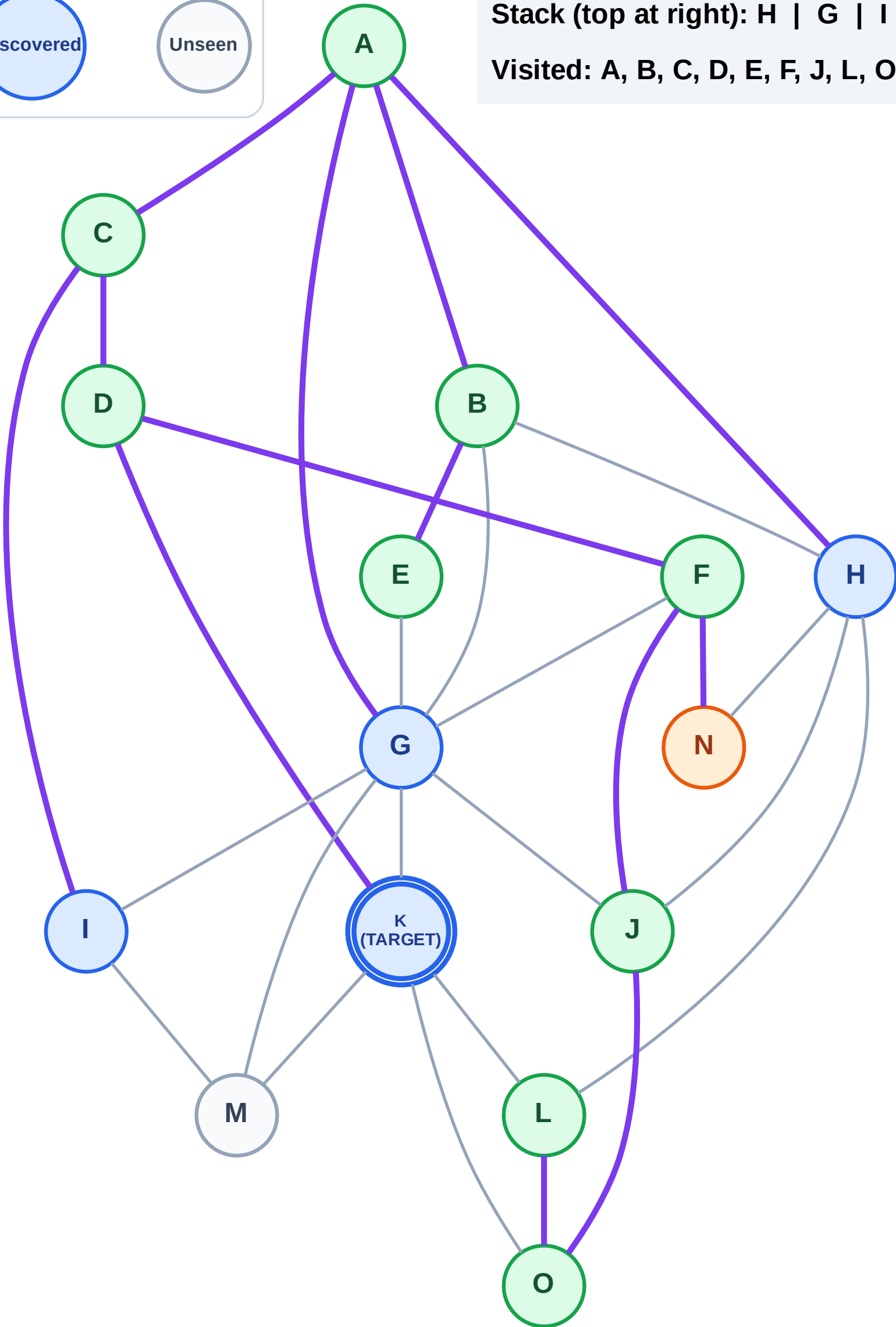
Processing

Discovered

Unseen

Stack (top at right): H | G | I | K

Visited: A, B, C, D, E, F, J, L, O



# DFS Traversal

Step 21: No new neighbors from N

## Legend



Complete



Processing



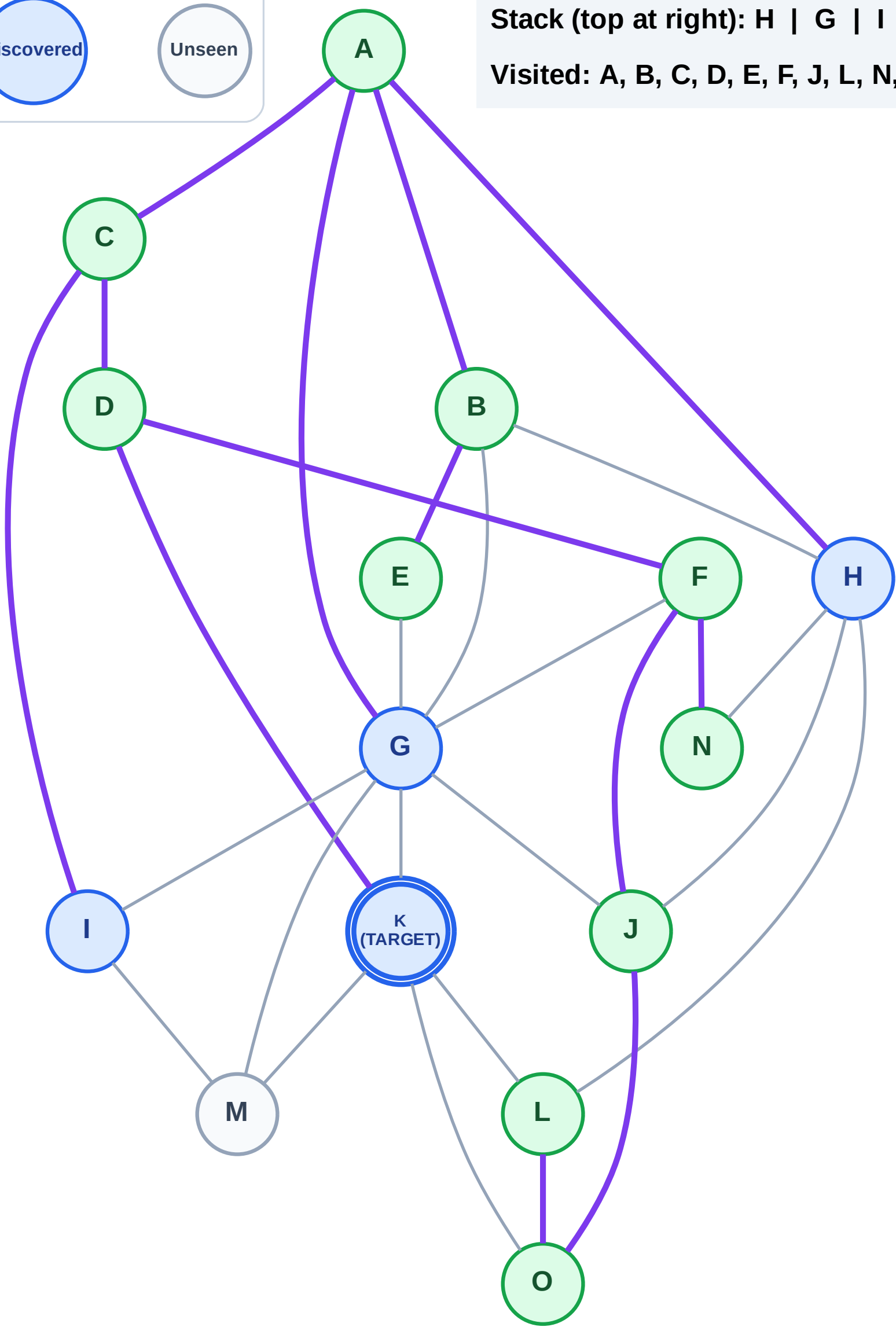
Discovered



Unseen

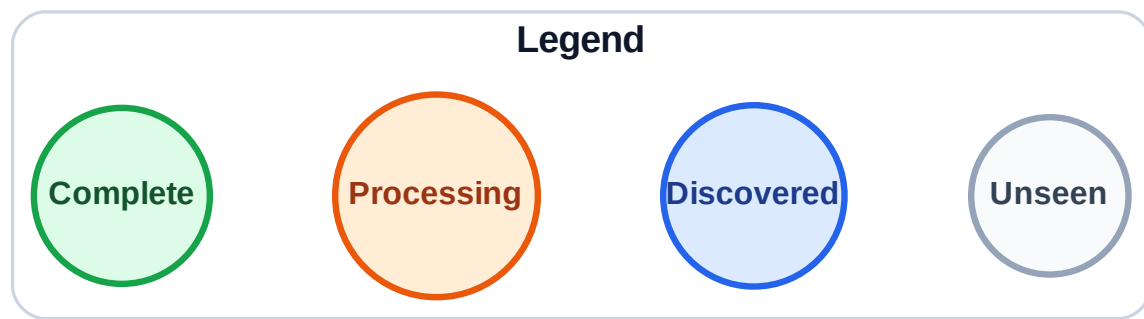
Stack (top at right): H | G | I | K

Visited: A, B, C, D, E, F, J, L, N, O

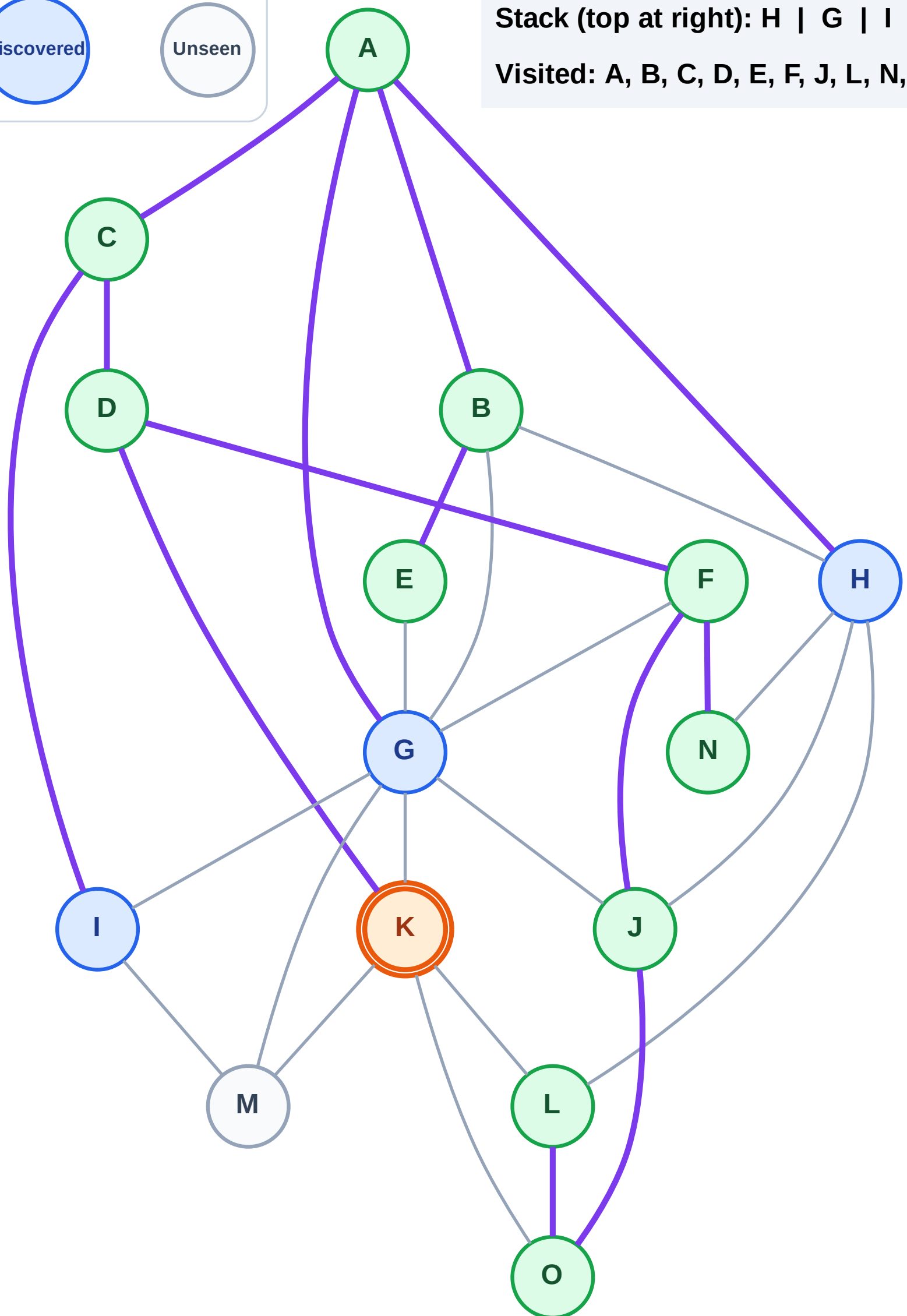


# DFS Traversal

Step 22: Process node K



Stack (top at right): H | G | I  
Visited: A, B, C, D, E, F, J, L, N, O



# DFS Traversal

Step 23: TARGET NODE K REACHED! Path (3 hops): A → C → D → K

## Legend



Stack (top at right): H | G | I

Visited: A, B, C, D, E, F, J, K, L, N, O

